MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE EML

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1575 failures per million operating hours for class I or 0.2835 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EML.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/277-1 for class I and M28787/277-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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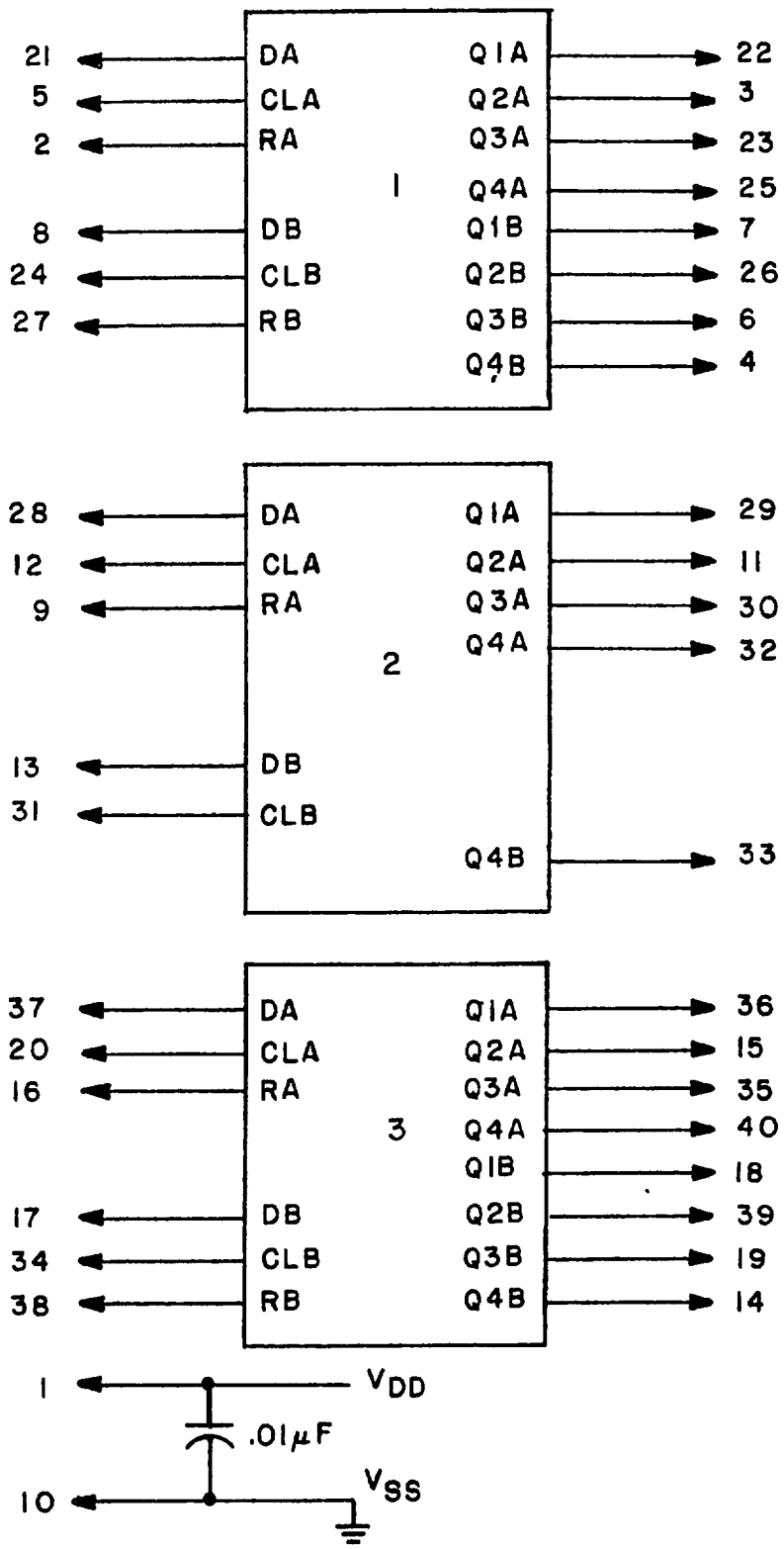


FIGURE 1. EML module function diagram, shift register.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{DD} stress	IV	Q	← Module must pass all remaining tests →										
C1-C17	Input clamp diode voltage, negative	V	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
D1-D17	Input clamp diode voltage, positive	VI	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
E1-E8	Functional	VII	X	← See table VII →										
F1-F21	Output voltage, low level, I_{OL} = 0	VIII	X	0	50	0	50	0	50	0	50	0	50	mV
G1-G21	Output voltage, high level, I_{OH} = 0	IX	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
H1-H21	Output voltage, low level, V_{DD} = 5 V	X	X	0	400	0	400	0	400	0	400	0	400	mV
J1-J21	Output voltage, high level, V_{DD} = 5 V	XI	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
K1-K21	Output voltage, low level, V_{DD} = 15 V	XII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
L1-L21	Output voltage, high level, V_{DD} = 15 V	XIII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
M1-M17	Input current, low level	XIV	X	0	-200	---	---	0	-1000	---	---	0	-1000	nA
N1-N17	Input current, high level	XV	X	0	200	---	---	0	1000	---	---	0	1000	nA
P1	Power supply current, low level	XVI	X	0	15	---	---	0	270	---	---	0	450	µA
P2	Power supply current, high level	XVI	X	0	15	---	---	0	270	---	---	0	450	µA
Q1-Q21	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	36	360	36	360	40	400	27	275	44	440	ns
Q22-Q42	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	96	960	96	960	115	1150	85	850	110	1100	ns
R1-R21	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XVIII	S	20	200	20	200	21	213	17	171	22	218	ns
R22-R42	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XVIII	Q	50	500	50	500	54	543	40	403	57	570	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S21	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XIX	S	75	750	75	750	76	764	72	720	77	773	ns
S22-S42	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XIX	Q	99	998	99	998	103	1038	91	910	105	1050	ns
T1-T21	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XX	S	75	750	75	750	76	764	72	720	77	773	ns
T22-T42	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	99	998	99	998	103	1038	91	910	105	1050	ns
U1-U20	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	75	750	75	750	76	764	72	720	77	773	ns
U21-U40	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	99	998	99	998	103	1038	91	910	105	1050	ns
V1-V21	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XXII	Q	18	180	18	180	19	191	15	156	20	198	ns
W1-W21	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	10	100	10	100	10	105	8	88	11	108	ns
X1-X21	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXIV	Q	37	375	37	375	38	383	35	359	38	387	ns
Y1-Y21	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DL} = 10 V	XXV	Q	37	375	37	375	38	383	35	359	38	387	ns
Z1-Z20	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXVI	Q	37	375	37	375	38	383	35	359	38	387	ns
AA1-AA21	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXVII	Q	13	130	13	130	13	138	11	112	14	143	ns
BB1-BB21	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXVIII	Q	8	80	8	80	8	84	7	70	8	87	ns
CC1-CC21	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIX	Q	17	170	17	170	17	174	16	160	17	177	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
DD1-DD21	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	17	170	17	170	17	174	16	160	17	177	ns
EE1-EE20	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXI	Q	17	170	17	170	17	174	16	160	17	177	ns
FF1-FF21	Setup time (t_{SL} and t_{SH})	XXXII	Q	2.3	3.7	2.3	3.7	2.3	3.7	2.3	3.7	2.3	3.7	µs
GG1-GG21	Maximum clock frequency check and minimum pulse width	XXXIII	S	1900	2300	1900	2300	1900	2300	1900	2300	1900	2300	ns
HH	Power supply filter capacitance	XXXIV	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
JJ1-JJ17	Input capacitance	XXXV	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through JJ shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through JJ in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through HH shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition HH, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through HH shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through JJ shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	18 max	18 max	mW
Input voltage			
Low level (V_{DD} = 5 V)	0 min, 1.5 max	0 min, 1.5 max	V
Low level (V_{DD} = 15 V)	0 min, 4 max	0 min, 4 max	V
High level (V_{DD} = 5 V)	3.5 min, V_{DD}max	3.5 min, V_{DD} max	V
High level (V_{DD} = 15 V)	11 min, V_{DD}max	11 min, V_{DD} max	V

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	Class I	Class II	Unit
Input current			
Low level (V_{IL} = 0 V, V_{DD} = 18 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V, V_{DD} = 18 V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5 V)			
Low level (V_{OL} = 0.4 V)			
I_{OL} at -55°C	---	0.64 min	mA
0°C and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
85°C	---	0.39 min	mA
High level (V_{OH} = 4.6 V)			
I_{OH} at -55°C	---	-0.64 min	mA
0°C and 25°C	-0.51 min	-0.51 min	mA
60°C	-0.44 min	-0.44 min	mA
85°C	---	-0.39 min	mA
Clock input, V_{DD} = 5 V			
Frequency	2 max	2 max	MHz
Pulse width	300 min	300 min	ns
Set, reset pulse width	350 min	350 min	ns

6.5 Module function. The module contains three independent dual four bit shift registers. Device 2B has the reset tied to V_{SS} and only the Q4B output is available.

6.5.1 Input to output relationship. Each register shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Inputs			Outputs	
Clock	D	R	Q1	Qn
↑	L	L	L	Qn-1
↑	H	L	H	Qn-1
↓	X	L	Q1 (no change)	Qn (no change)
X	X	H	L	L

1/ H = High voltage level
2/ L = Low voltage level
3/ X = Don't care

6.5.2 Paralleling of outputs. Paralleling of outputs is not permissible.

6.5.3 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{TOTAL} = (C\ V_{DD}^2 f)\ N + \text{Quiescent power dissipation.}$$

Where: C = load capacitance.
V_{DD} = Supply voltage.
f = operating frequency.
N = number of outputs.

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6.6. Module handling consideration. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material except when being tested or in actual operation to avoid build-up of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

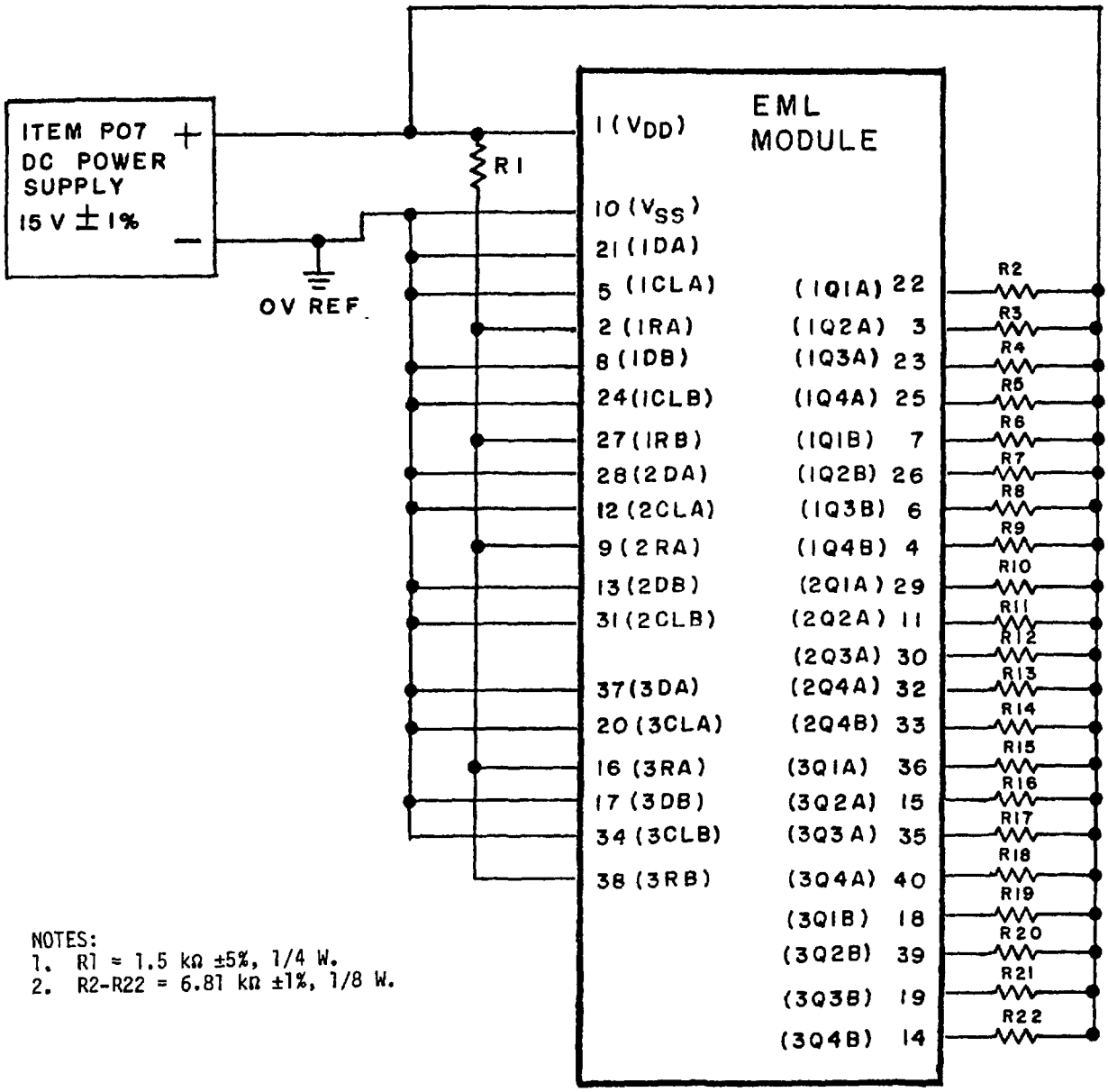
e. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

Agent:
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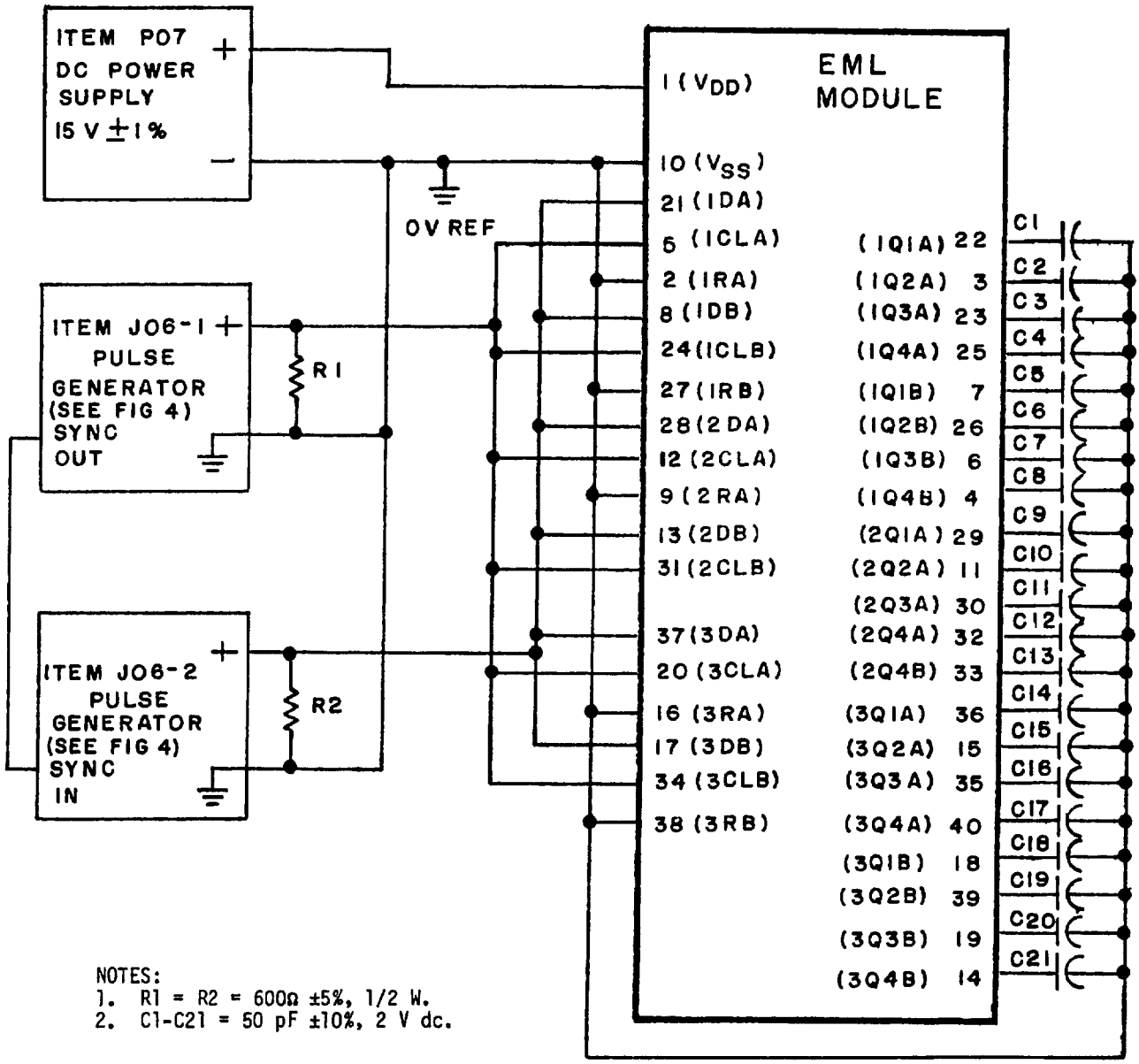
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NOTES:
1. R1 = 1.5 kΩ ±5%, 1/4 W.
2. R2-R22 = 6.81 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

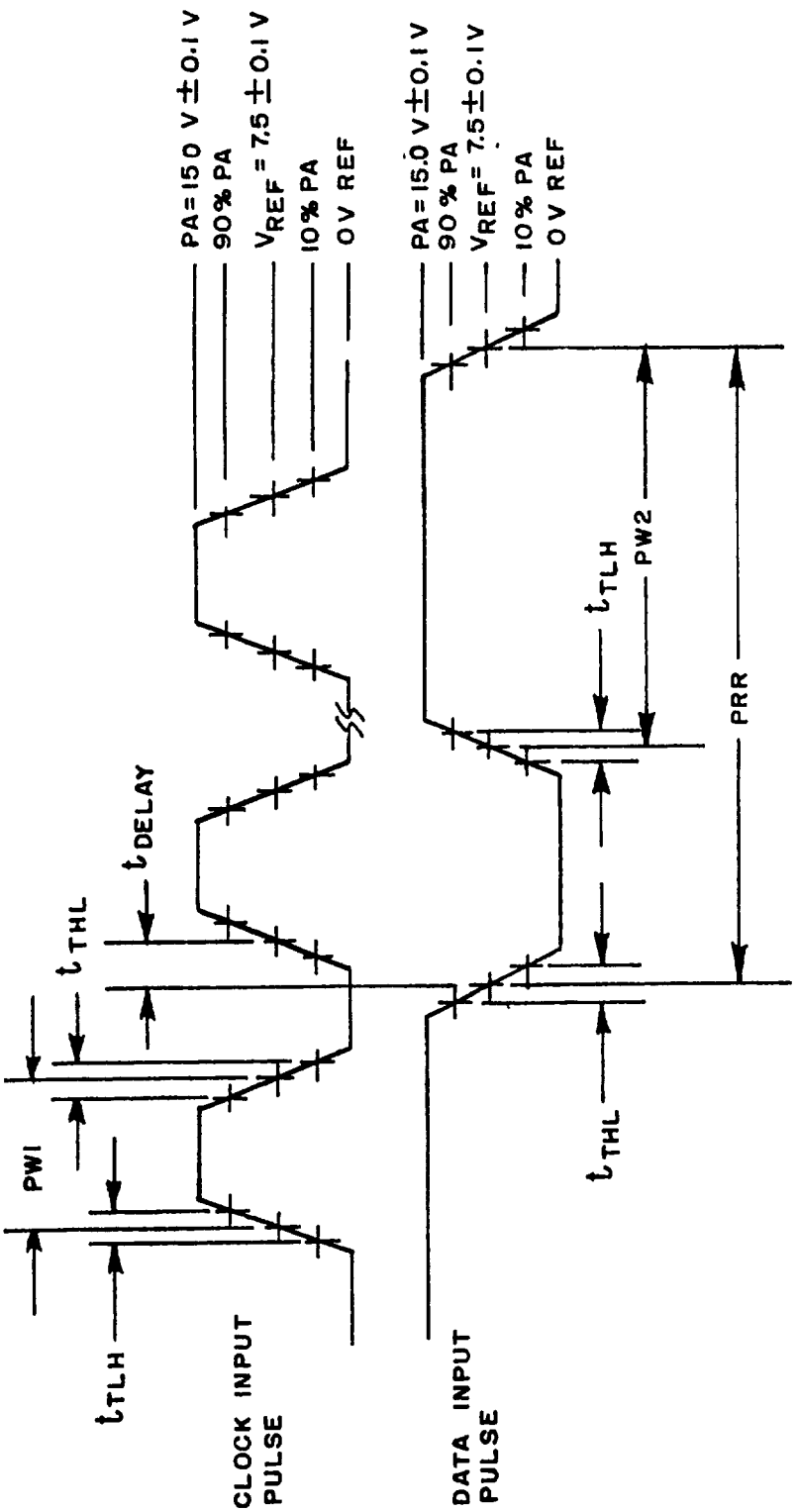
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NOTES:
1. R1 = R2 = 600Ω ±5%, 1/2 W.
2. C1-C21 = 50 pF ±10%, 2 V dc.

FIGURE 3. Component temperature test circuit.

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NOTES:
1. Item J06-1 parameters (clock input pulse):
   PW1 = 500 ±50 ns.
   t_{TLH} = t_{THL} = 50 ±10 ns.
   Double pulse mode.
2. Item J06-2 parameters (data input pulse):
   PW2 = 1.0 ±0.1 µs.
   t_{TLH} = t_{THL} = 50 ±10 ns.
   PRP = 2.0 ±0.2 µs.
3. t_{DELAY} = 600 ±60 ns.

FIGURE 4. Waveform setup (component temperature test) for figure 3.

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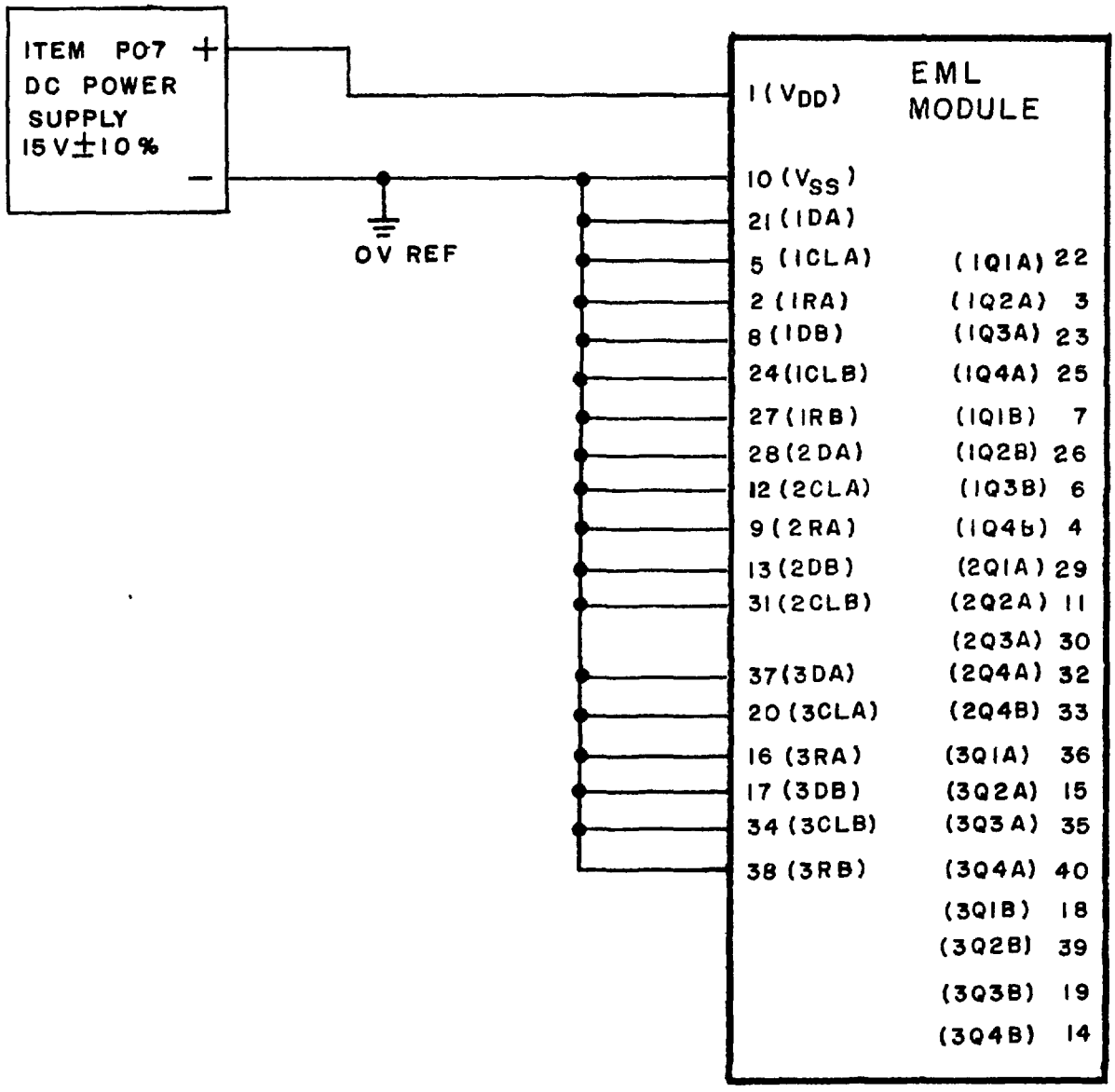


FIGURE 5. Vibration test circuit.

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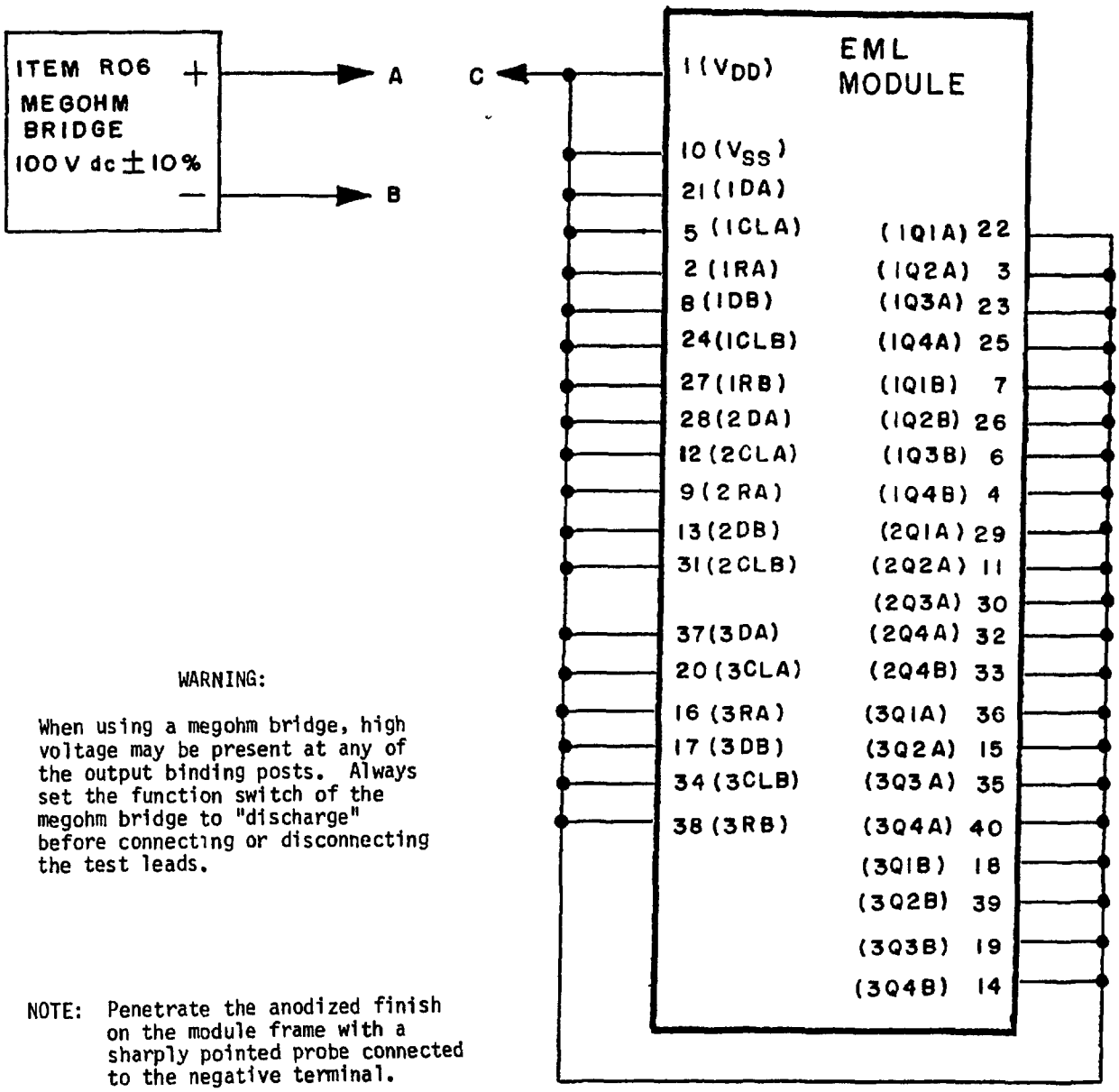
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

TABLE IV. V_{DD} stress test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
B	V_{DD} stress	Power supply voltage for a period of 5.0 ± 0.1s	Module must pass all remaining tests			7

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table III.

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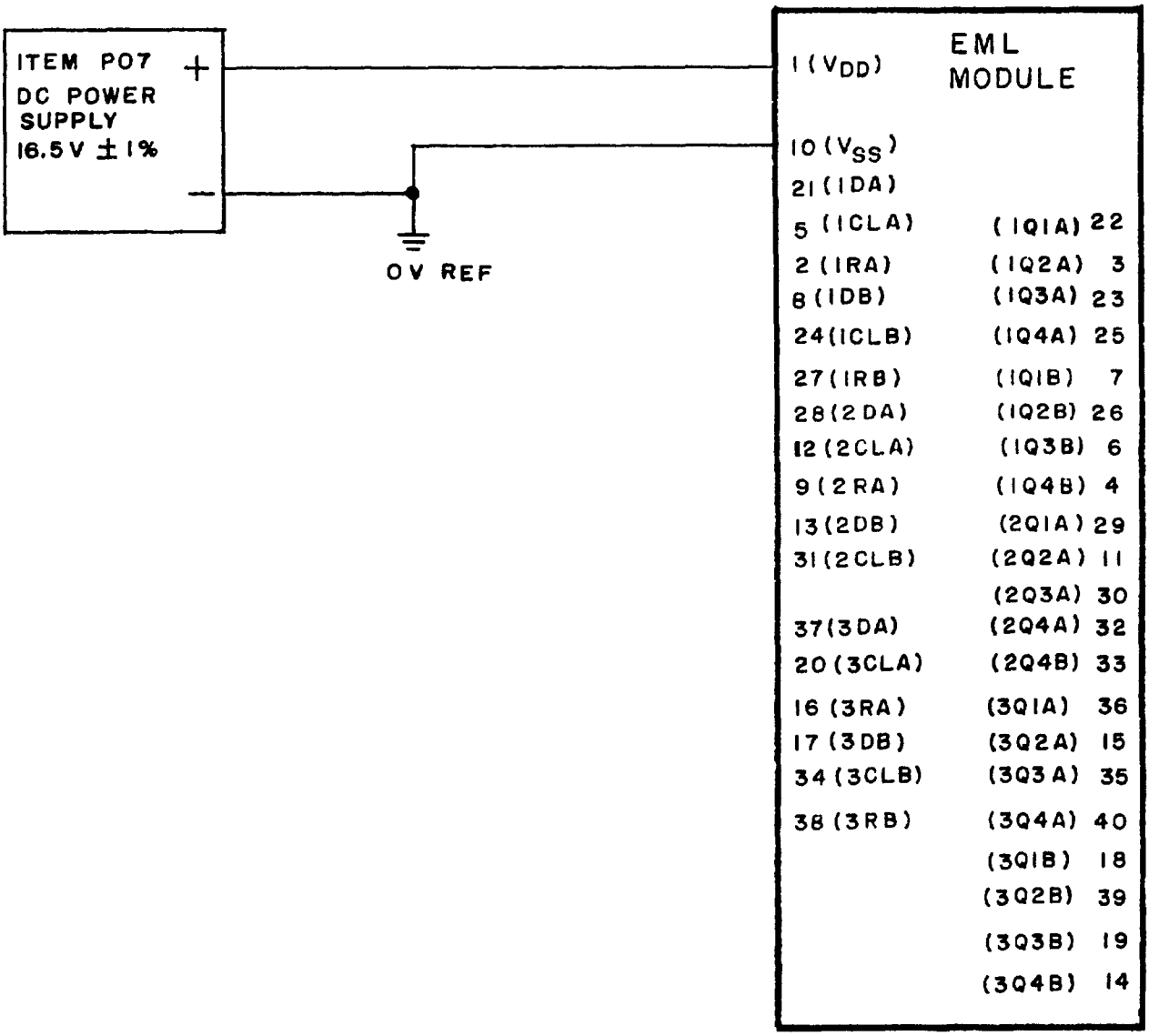


FIGURE 7. V_{DD} stress test circuit for table IV.

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TABLE V. Input clamp diode voltage (negative) test.

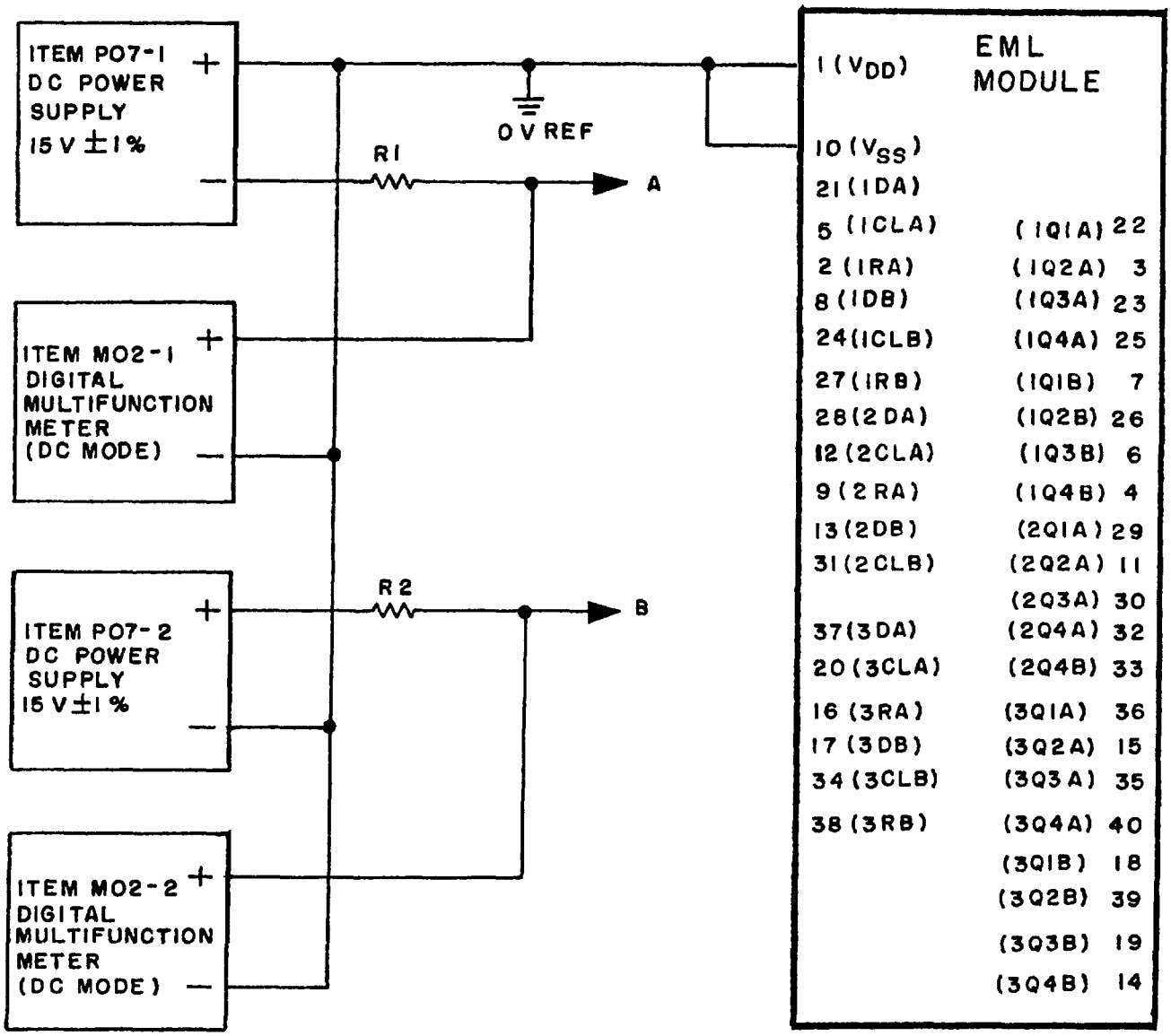
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1DA Input clamp diode voltage, negative	21	Open	-0.4	-6.0	V	8
C2	1CLA "	5	"	"	"	"	"
C3	1RA "	2	"	"	"	"	"
C4	1DB "	8	"	"	"	"	"
C5	1CLB "	24	"	"	"	"	"
C6	1RB "	27	"	"	"	"	"
C7	2DA "	28	"	"	"	"	"
C8	2CLA "	12	"	"	"	"	"
C9	2RA "	9	"	"	"	"	"
C10	2DB "	13	"	"	"	"	"
C11	2CLB "	31	"	"	"	"	"
C12	3DA "	37	"	"	"	"	"
C13	3CLA "	20	"	"	"	"	"
C14	3RA "	16	"	"	"	"	"
C15	3DB "	17	"	"	"	"	"
C16	3CLB "	34	"	"	"	"	"
C17	3RB "	38	"	"	"	"	"

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TABLE VI. Input clamp diode voltage (positive) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	1DA	Input clamp diode voltage, positive	Open	21	0.4	1.5	V	8
D2	1CLA	"	"	5	"	"	"	"
D3	1RA	"	"	2	"	"	"	"
D4	1DB	"	"	8	"	"	"	"
D5	1CLB	"	"	24	"	"	"	"
D6	1RB	"	"	27	"	"	"	"
D7	2DA	"	"	28	"	"	"	"
D8	2CLA	"	"	12	"	"	"	"
D9	2RA	"	"	9	"	"	"	"
D10	2DB	"	"	13	"	"	"	"
D11	2CLB	"	"	31	"	"	"	"
D12	3DA	"	"	37	"	"	"	"
D13	3CLA	"	"	20	"	"	"	"
D14	3RA	"	"	16	"	"	"	"
D15	3DB	"	"	17	"	"	"	"
D16	3CLB	"	"	34	"	"	"	"
D17	3RB	"	"	38	"	"	"	"

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1, P07-2, M02-1, and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 8. Input clamp diode voltage (negative and positive) test circuit for tables V and VI.

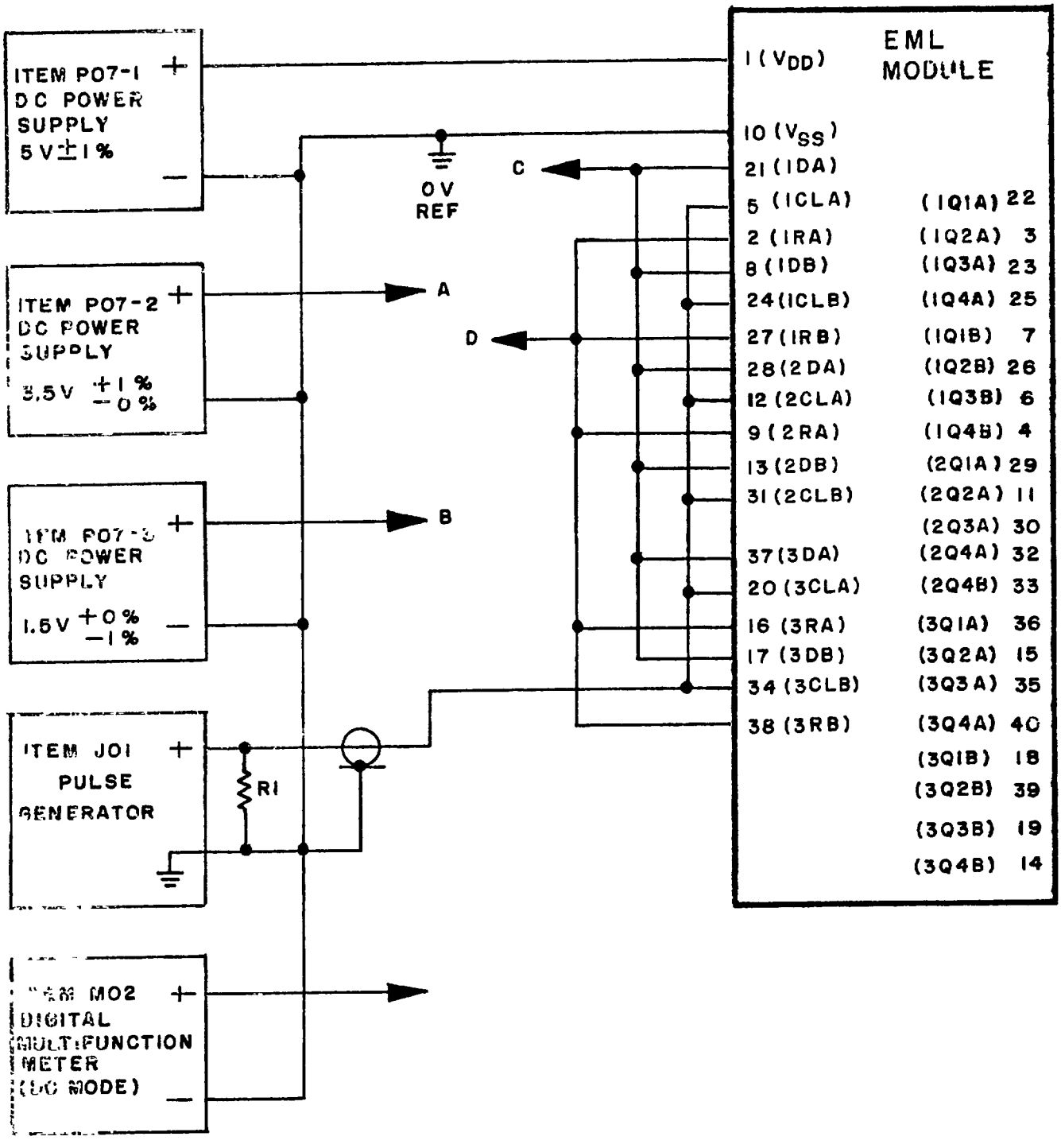
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TABLE VII. Functional test.

Test condition	Parameter	Connect 1/		Output pins																						Initial limits 25°C ± 5°C ambient		Test figure
		A to pin	B to pin	22	7	29	36	18	3	26	11	15	39	23	6	30	35	19	25	4	32	33	40	14		Min	Max	
E1	Functional	D	C	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	X	L	L		2/-4/	2/-4/	9
E2	"	C	D	H	H	H	H	H	L	L	L	L	L	L	L	L	L	L	L	L	L	X	L	L		"	"	"
E3	"	Open	C,D	L	L	L	L	L	H	H	H	H	H	L	L	L	L	L	L	L	L	X	L	L		"	"	"
E4	"	C	D	H	H	H	H	H	L	L	L	L	L	H	H	H	H	H	L	L	L	L	L	L		"	"	"
E5	"	"	"	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	H	H	H	H	H	H		"	"	"
E6	"	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"	"	"
E7	"	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H		"	"	"
E8	"	D	C	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L		"	"	"

1/ Apply one pulse using item J01 after establishing input conditions for test conditions E1 through E8.
2/ L ≤ 0.4 V.
3/ H ≥ 4.6 V.
4/ X = Don't care.

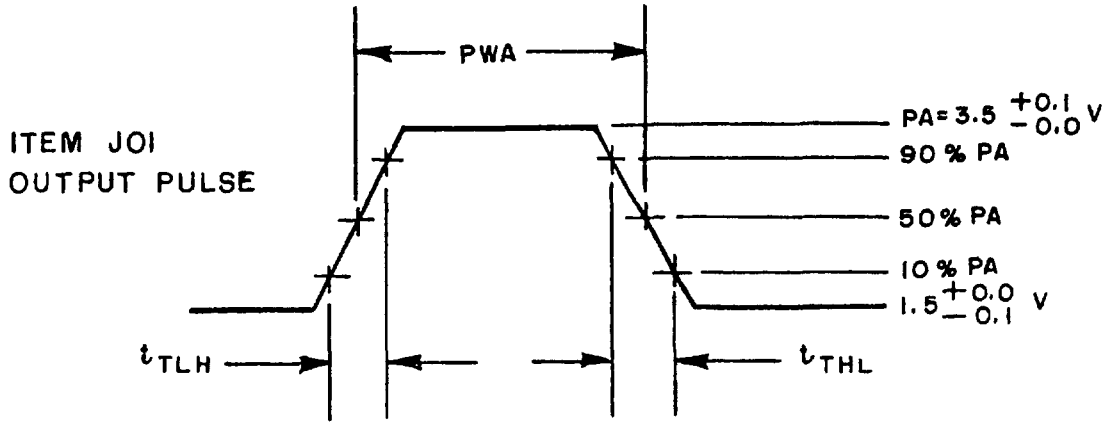
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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. Verify that the output high level is ≥ 4.6 V and the low level is ≤ 0.4 V using item M02.

FIGURE 9. Functional test circuit for table VII.

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NOTE: Input pulse parameters:
PWA = 1.0 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
Single pulse mode.

FIGURE 9. Functional test circuit for table VII - Continued.

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TABLE VIII. Output voltage (low-level, I_{OL} = 0) test.

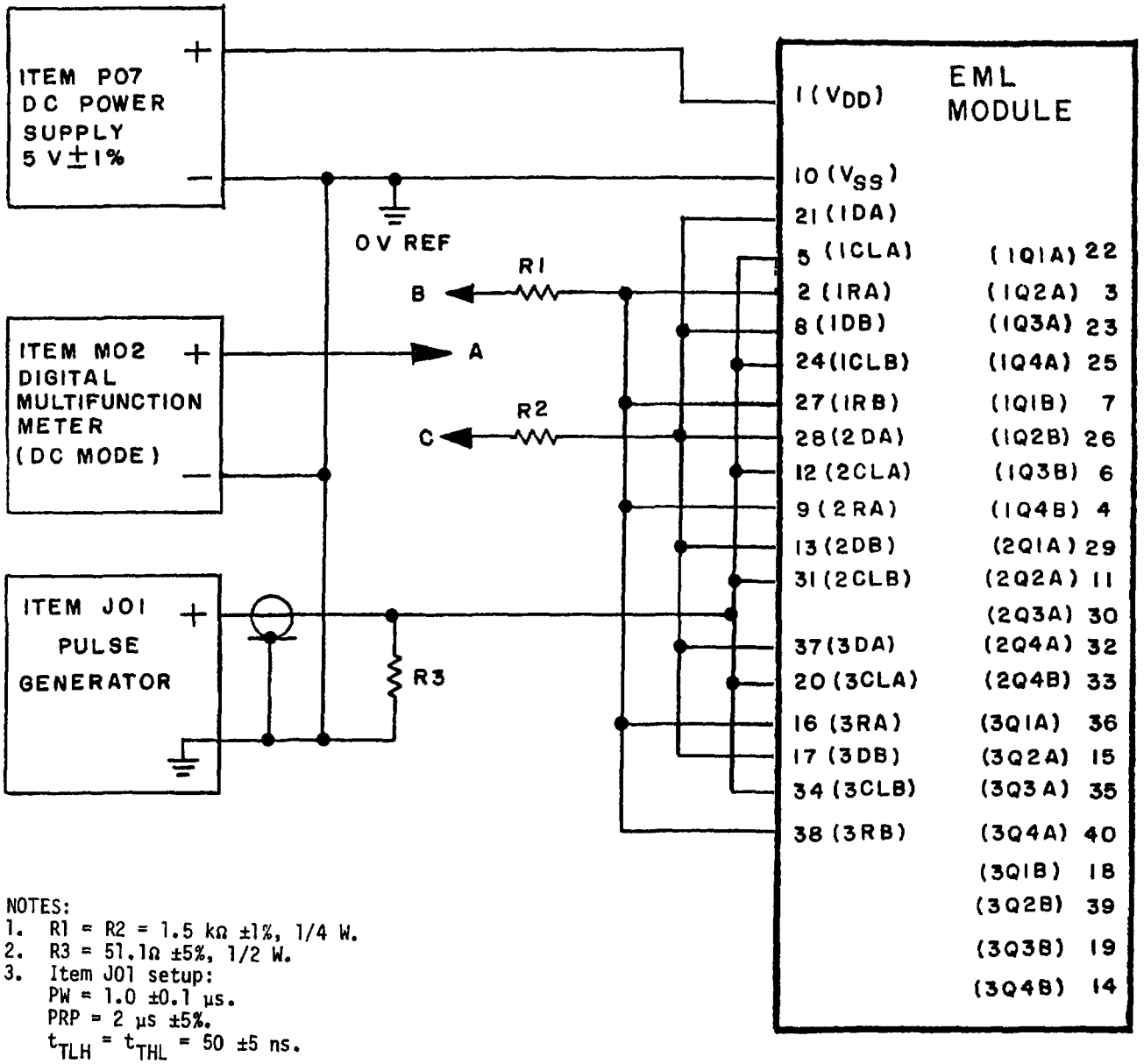
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1Q1A Output voltage, low level, I_{OL} = 0	22	10	10	0	50	mV	10
F2	1Q2A "	3	"	"	"	"	"	"
F3	1Q3A "	23	"	"	"	"	"	"
F4	1Q4A "	25	"	"	"	"	"	"
F5	1Q1B "	7	"	"	"	"	"	"
F6	1Q2B "	26	"	"	"	"	"	"
F7	1Q3B "	6	"	"	"	"	"	"
F8	1Q4B "	4	"	"	"	"	"	"
F9	2Q1A "	29	"	"	"	"	"	"
F10	2Q2A "	11	"	"	"	"	"	"
F11	2Q3A "	30	"	"	"	"	"	"
F12	2Q4A "	32	"	"	"	"	"	"
F13	2Q4B "	33	"	"	"	"	"	"
F14	3Q1A "	36	"	"	"	"	"	"
F15	3Q2A "	15	"	"	"	"	"	"
F16	3Q3A "	35	"	"	"	"	"	"
F17	3Q4A "	40	"	"	"	"	"	"
F18	3Q1B "	18	"	"	"	"	"	"
F19	3Q2B "	39	"	"	"	"	"	"
F20	3Q3B "	19	"	"	"	"	"	"
F21	3Q4B "	14	"	"	"	"	"	"

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TABLE IX. Output voltage (high-level, I_{OH} = 0) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1Q1A Output voltage, high level, I_{OH} = 0	22	10	1	4.95	5.05	V	10
G2	1Q2A "	3	"	"	"	"	"	"
G3	1Q3A "	23	"	"	"	"	"	"
G4	1Q4A "	25	"	"	"	"	"	"
G5	1Q1B "	7	"	"	"	"	"	"
G6	1Q2B "	26	"	"	"	"	"	"
G7	1Q3B "	6	"	"	"	"	"	"
G8	1Q4B "	4	"	"	"	"	"	"
G9	2Q1A "	29	"	"	"	"	"	"
G10	2Q2A "	11	"	"	"	"	"	"
G11	2Q3A "	30	"	"	"	"	"	"
G12	2Q4A "	32	"	"	"	"	"	"
G13	2Q4B "	33	"	"	"	"	"	"
G14	3Q1A "	36	"	"	"	"	"	"
G15	3Q2A "	15	"	"	"	"	"	"
G16	3Q3A "	35	"	"	"	"	"	"
G17	3Q4A "	40	"	"	"	"	"	"
G18	3Q1B "	18	"	"	"	"	"	"
G19	3Q2B "	39	"	"	"	"	"	"
G20	3Q3B "	19	"	"	"	"	"	"
G21	3Q4B "	14	"	"	"	"	"	"

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NOTES:
1. R1 = R2 = 1.5 kΩ ±1%, 1/4 W.
2. R3 = 51.1Ω ±5%, 1/2 W.
3. Item J01 setup:
   PW = 1.0 ±0.1 µs.
   PRP = 2 µs ±5%.
   $t_{TLH} = t_{THL}$ = 50 ±5 ns.
   PA = 5 V ±1%.
4. Setup item J01, then put in single pulse mode with connections made according to tables VIII and IX. Apply four pulses before making measurements.

FIGURE 10. Output voltage (low-level and high-level) test circuit for tables VIII and IX.

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TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	21,8,28,13,37,17	Open	Open	Open	---	---	---	11
H1	1Q1A Output voltage, low level, V_{DD} = 5 V	8,28,13,37,17	21	"	22	0	400	mV	"
H2	1Q2A "	21,8,28,13,37,17	Open	"	3	"	"	"	"
H3	1Q3A "	"	"	"	23	"	"	"	"
H4	1Q4A "	"	"	"	25	"	"	"	"
H5	1Q1B "	21,28,13,37,17	8	"	7	"	"	"	"
H6	1Q2B "	21,8,28,13,37,17	Open	"	26	"	"	"	"
H7	1Q3B "	"	"	"	6	"	"	"	"
H8	1Q4B "	"	"	"	4	"	"	"	"
H9	2Q1A "	21,8,13,37,17	28	"	29	"	"	"	"
H10	2Q2A "	21,8,28,37,17	13	"	11	"	"	"	"
H11	2Q3A "	21,8,28,13,37,17	Open	"	30	"	"	"	"
H12	2Q4A "	"	"	"	32	"	"	"	"
H13	2Q4B "	"	"	"	33	"	"	"	"
H14	3Q1A "	21,8,28,13,17	37	"	36	"	"	"	"
H15	3Q2A "	21,8,28,13,37,17	Open	"	15	"	"	"	"
H16	3Q3A "	"	"	"	35	"	"	"	"
H17	3Q4A "	"	"	"	40	"	"	"	"
H18	3Q1B "	21,8,28,13,37	17	"	18	"	"	"	"
H19	3Q2B "	21,8,28,13,37,17	Open	"	39	"	"	"	"
H20	3Q3B "	"	"	"	19	"	"	"	"
H21	3Q4B "	"	"	"	14	"	"	"	"

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TABLE XI. Output voltage (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup		Open	21,8,28,13,37,17	---	---	---	---	---	11
J1	1Q1A	Output voltage, high level, V_{DD} = 5 V	21	8,28,13,37,17	22	Open	4.6	5.0	V	"
J2	1Q2A	"	Open	21,8,28,13,37,17	3	"	"	"	"	"
J3	1Q3A	"	"	"	23	"	"	"	"	"
J4	1Q4A	"	"	"	25	"	"	"	"	"
J5	1Q1B	"	8	21,28,13,37,17	7	"	"	"	"	"
J6	1Q2B	"	Open	21,8,28,13,37,17	26	"	"	"	"	"
J7	1Q3B	"	"	"	6	"	"	"	"	"
J8	1Q4B	"	"	"	4	"	"	"	"	"
J9	2Q1A	"	28	21,8,13,37,17	29	"	"	"	"	"
J10	2Q2A	"	13	21,8,28,37,17	11	"	"	"	"	"
J11	2Q3A	"	Open	21,8,28,13,37,17	30	"	"	"	"	"
J12	2Q4A	"	"	"	32	"	"	"	"	"
J13	2Q4B	"	"	"	33	"	"	"	"	"
J14	3Q1A	"	37	21,8,28,13,17	36	"	"	"	"	"
J15	3Q2A	"	Open	21,8,28,13,37,17	15	"	"	"	"	"
J16	3Q3A	"	"	"	35	"	"	"	"	"
J17	3Q4A	"	"	"	40	"	"	"	"	"
J18	3Q1B	"	17	21,8,28,13,37	18	"	"	"	"	"
J19	3Q2B	"	Open	21,8,28,13,37,17	39	"	"	"	"	"
J20	3Q3B	"	"	"	19	"	"	"	"	"
J21	3Q4B	"	"	"	14	"	"	"	"	"

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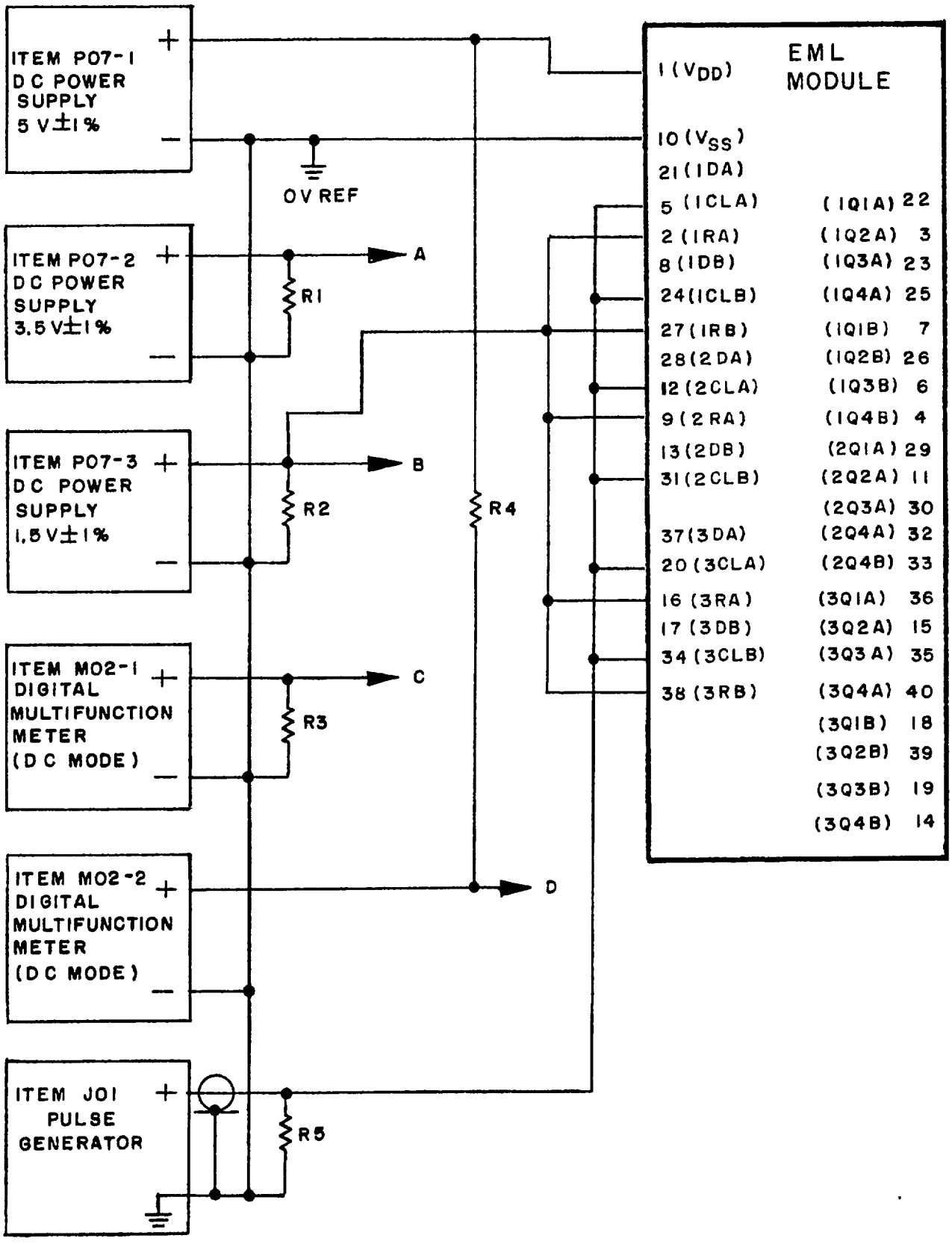


FIGURE 11. Output voltage (low-level and high-level) test circuit for tables X and XI.

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NOTES:
1. R1 = R2 = R5 = 51.1Ω ±5%, 1/8 W.
2. R3 = R4 = 9.09 kΩ ±1%, 1/8 W at 0°C and 25°C.
   = 10.5 kΩ ±1%, 1/8 W at 60°C.
   = 11 kΩ ±1%, 1/8 W at 85°C.
   = 7.15 kΩ ±1%, 1/8 W at -55°C.
3. Item J01 setup:
   PA = 5 V ±1%.
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns.
   Single pulse mode.
4. After establishing the setup input condition, apply four pulses using item J01.
5. After establishing the input conditions of test conditions F1 or G1, apply one pulse using item J01. Prior to making the specified measurement, repeat this sequence for the remainder of the test.

FIGURE 11. Output voltage (low-level and high-level) test circuit for tables X and XI - Continued.

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TABLE XII. Output voltage (low-level) test.

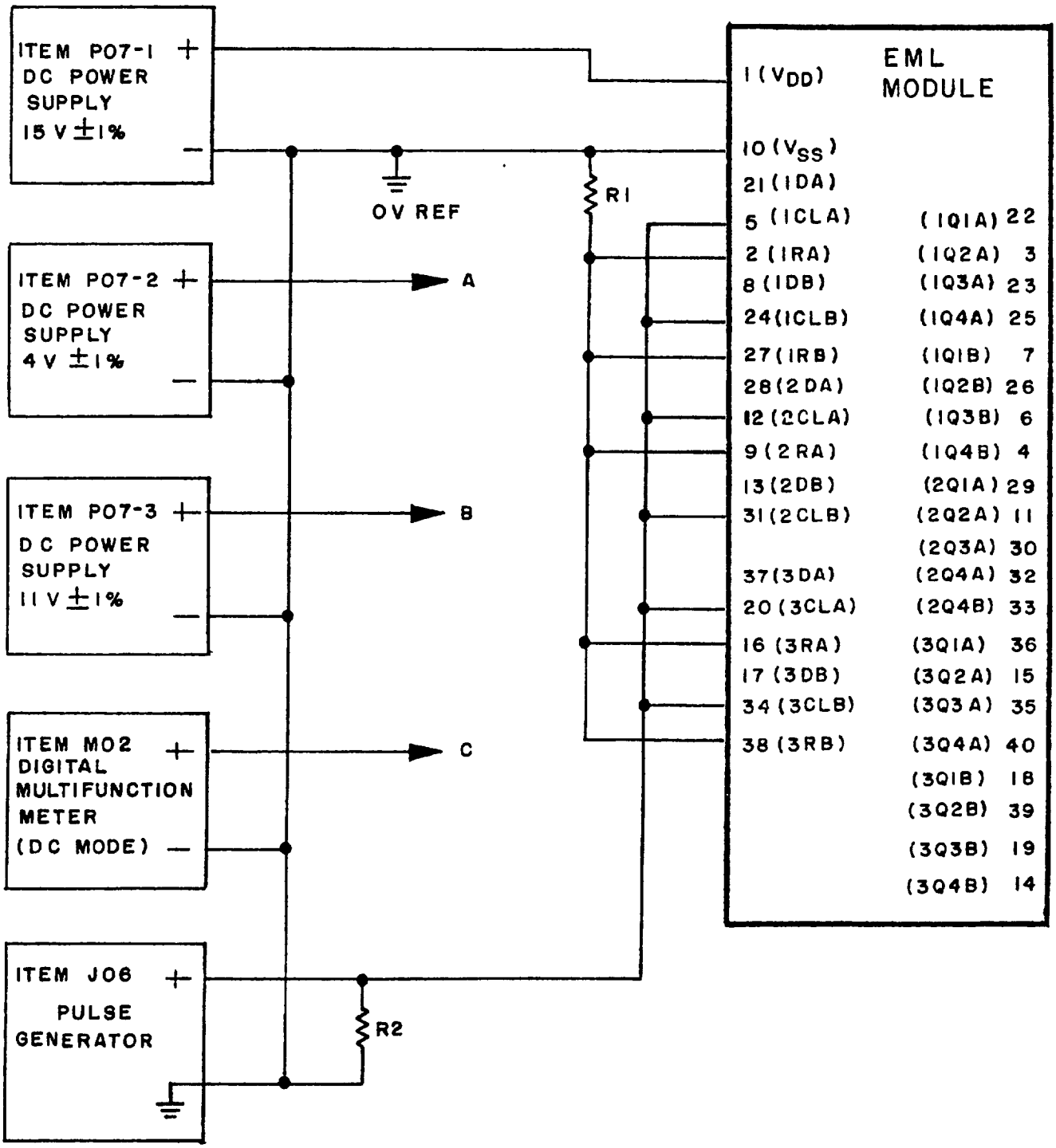
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
---	Setup	21,8,28,13,37,17	Open	---	---	---	---	12
K1	1Q1A Output voltage, low level, V_{DD} = 15 V	"	"	22	0.0	1.5	V	"
K2	1Q2A "	"	"	3	"	"	"	"
K3	1Q3A "	"	"	23	"	"	"	"
K4	1Q4A "	"	"	25	"	"	"	"
K5	1Q1B "	"	"	7	"	"	"	"
K6	1Q2B "	"	"	26	"	"	"	"
K7	1Q3B "	"	"	6	"	"	"	"
K8	1Q4B "	"	"	4	"	"	"	"
K9	2Q1A "	"	"	29	"	"	"	"
K10	2Q2A "	"	"	11	"	"	"	"
K11	2Q3A "	"	"	30	"	"	"	"
K12	2Q4A "	"	"	32	"	"	"	"
K13	2Q4B "	"	"	33	"	"	"	"
K14	3Q1A "	"	"	36	"	"	"	"
K15	3Q2A "	"	"	15	"	"	"	"
K16	3Q3A "	"	"	35	"	"	"	"
K17	3Q4A "	"	"	40	"	"	"	"
K18	3Q1B "	"	"	18	"	"	"	"
K19	3Q2B "	"	"	39	"	"	"	"
K20	3Q3B "	"	"	19	"	"	"	"
K21	3Q4B "	"	"	14	"	"	"	"

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TABLE XIII. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
---	Setup	---	21,8,28,13,37,17	---	---	---	---	12
L1	1Q1A Output voltage, high level, V_{DD} = 15 V	Open	"	22	13.5	15.0	V	"
L2	1Q2A "	"	"	3	"	"	"	"
L3	1Q3A "	"	"	23	"	"	"	"
L4	1Q4A "	"	"	25	"	"	"	"
L5	1Q1B "	"	"	7	"	"	"	"
L6	1Q2B "	"	"	26	"	"	"	"
L7	1Q3B "	"	"	6	"	"	"	"
L8	1Q4B "	"	"	4	"	"	"	"
L9	2Q1A "	"	"	29	"	"	"	"
L10	2Q2A "	"	"	11	"	"	"	"
L11	2Q3A "	"	"	30	"	"	"	"
L12	2Q4A "	"	"	32	"	"	"	"
L13	2Q4B "	"	"	33	"	"	"	"
L14	3Q1A "	"	"	36	"	"	"	"
L15	3Q2A "	"	"	15	"	"	"	"
L16	3Q3A "	"	"	35	"	"	"	"
L17	3Q4A "	"	"	40	"	"	"	"
L18	3Q1B "	"	"	18	"	"	"	"
L19	3Q2B "	"	"	39	"	"	"	"
L20	3Q3B "	"	"	19	"	"	"	"
L21	3Q4B "	"	"	14	"	"	"	"

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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/4 W.
3. Item J06 setup:
   PA = 15 V ±1%.
   PW = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.2 µs.
   $t_{TLH} = t_{THL} = 20 \pm 2$ ns.
   Single pulse mode.
4. Using item J06, apply four pulses, after establishing the setup condition of tables XII and XIII.

FIGURE 12. Output voltage (low-level and high-level) test circuit for tables XII and XIII.

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TABLE XIV. Input current (low-level) test.

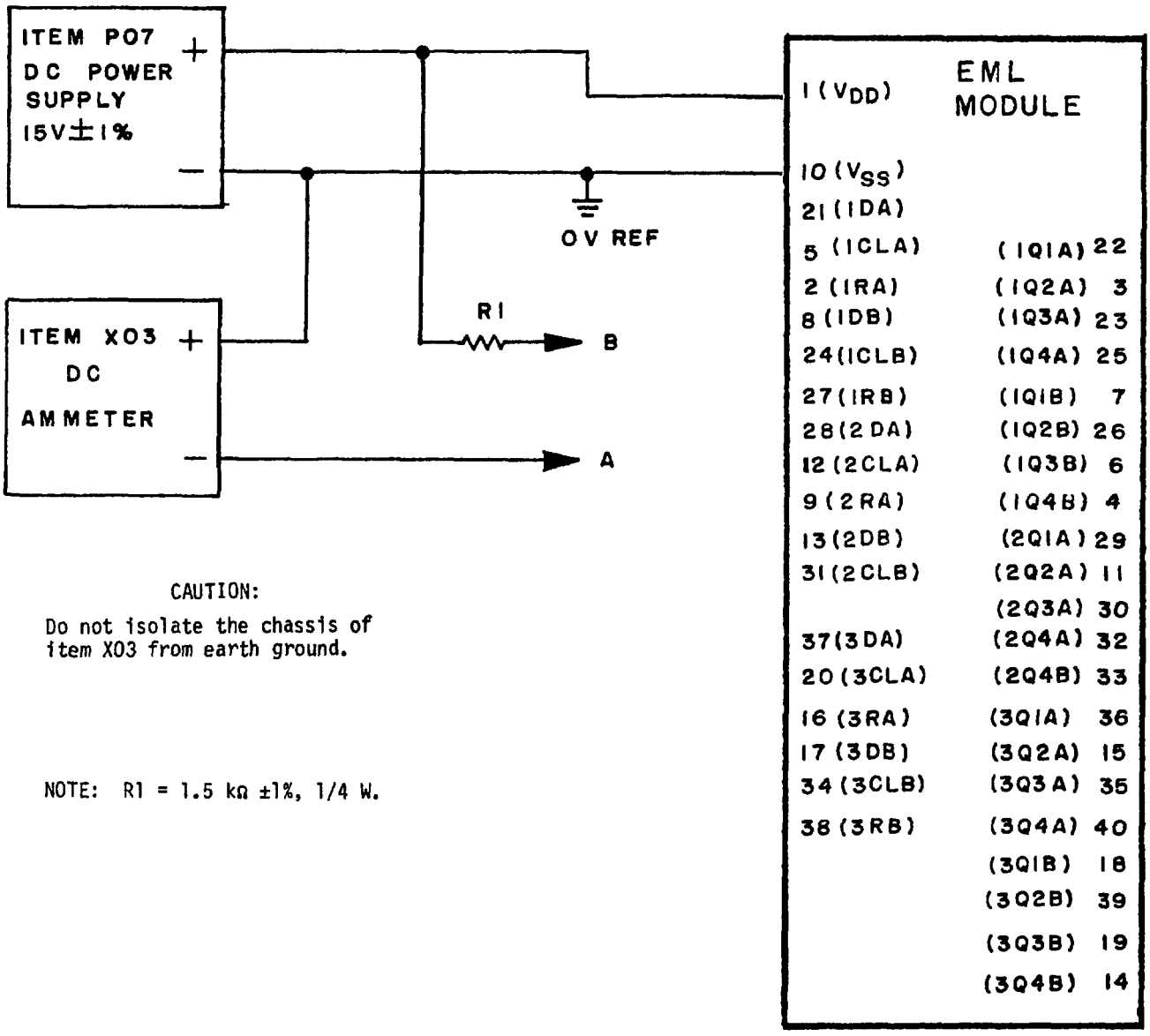
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
M1	1DA	Input current, low level	21	5,2,8,24,27,28,12,9,13, 31,37,20,16,17,34,38	0	-200	nA	13
M2	1CLA	"	5	21,2,8,24,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
M3	1RA	"	2	21,5,8,24,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
M4	1DB	"	8	21,5,2,24,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
M5	1CLB	"	24	21,5,2,8,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
M6	1RB	"	27	21,5,2,8,24,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
M7	2DA	"	28	21,5,2,8,24,27,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
M8	2CLA	"	12	21,5,2,8,24,27,28,9,13, 31,37,20,16,17,34,38	"	"	"	"
M9	2RA	"	9	21,5,2,8,24,27,28,12,13, 31,37,20,16,17,34,38	"	"	"	"
M10	2DB	"	13	21,5,2,8,24,27,28,12,9, 31,37,20,16,17,34,38	"	"	"	"
M11	2CLB	"	31	21,5,2,8,24,27,28,12,9, 13,37,20,16,17,34,38	"	"	"	"
M12	3DA	"	37	21,5,2,8,24,27,28,12,9, 13,31,20,16,17,34,38	"	"	"	"
M13	3CLA	"	20	21,5,2,8,24,27,28,12,9, 13,31,37,16,17,34,38	"	"	"	"
M14	3RA	"	16	21,5,2,8,24,27,28,12,9, 13,31,37,20,17,34,38	"	"	"	"

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TABLE XIV. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M15	3DB Input current, low level	17	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,34,38	0	-200	nA	13
M16	3CLB "	34	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,17,38	"	"	"	"
M17	3RB "	38	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,17,34	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO3 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 13. Input current (low-level) test circuit for table XIV.

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TABLE XV. Input current (high-level) test.

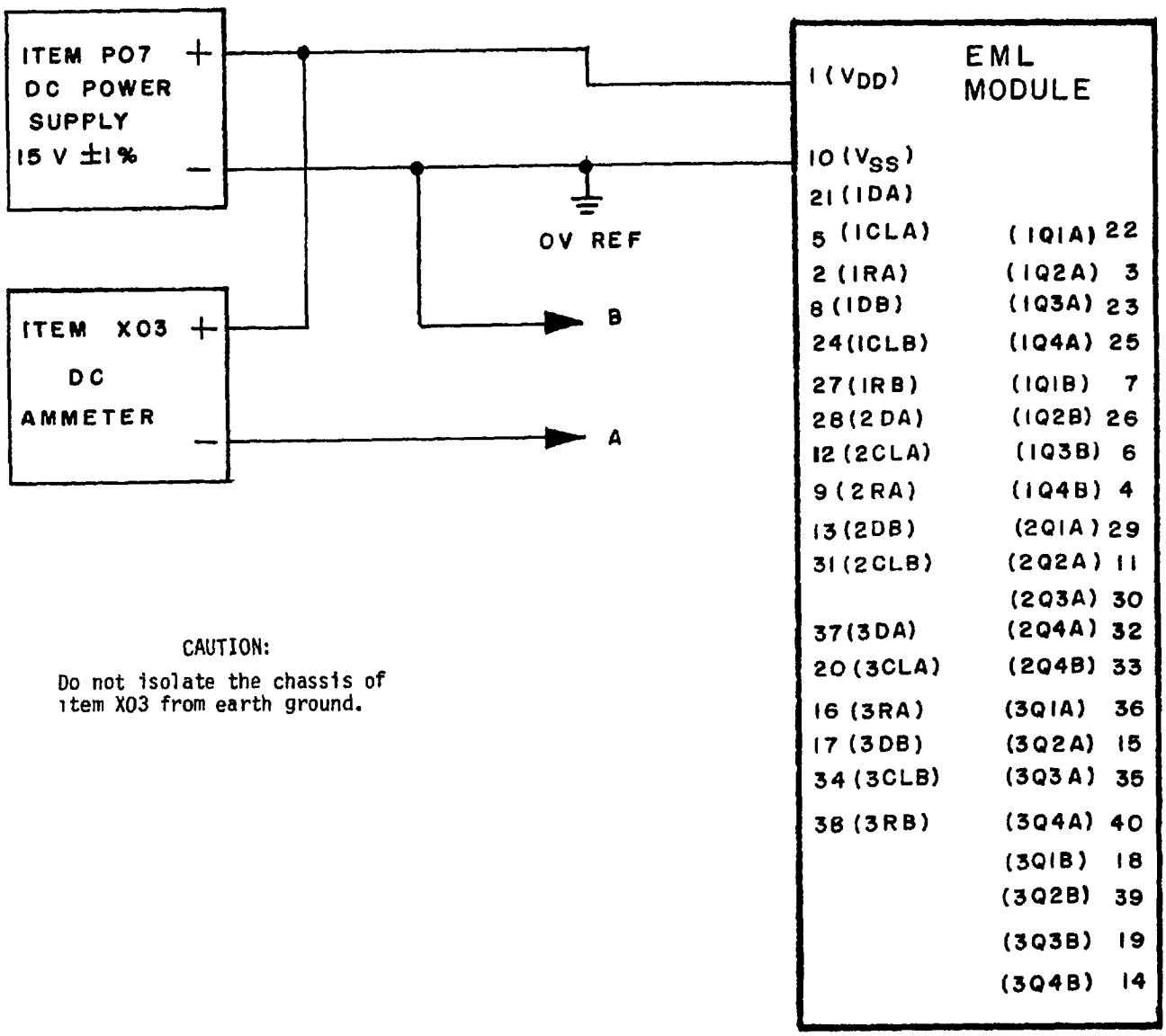
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N1	1DA	Input current, high level	21	5,2,8,24,27,28,12,9,13, 31,37,20,16,17,34,38	0	200	nA	14
N2	1CLA	"	5	21,2,8,24,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
N3	1RA	"	2	21,5,8,24,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
N4	1DB	"	8	21,5,2,24,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
N5	1CLB	"	24	21,5,2,8,27,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
N6	1RB	"	27	21,5,2,8,24,28,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
N7	2DA	"	28	21,5,2,8,24,27,12,9,13, 31,37,20,16,17,34,38	"	"	"	"
N8	2CLA	"	12	21,5,2,8,24,27,28,9,13, 31,37,20,16,17,34,38	"	"	"	"
N9	2RA	"	9	21,5,2,8,24,27,28,12,13, 31,37,20,16,17,34,38	"	"	"	"
N10	2DB	"	13	21,5,2,8,24,27,28,12,9, 31,37,20,16,17,34,38	"	"	"	"
N11	2CLB	"	31	21,5,2,8,24,27,28,12,9, 13,37,20,16,17,34,38	"	"	"	"
N12	3DA	"	37	21,5,2,8,24,27,28,12,9, 13,31,20,16,17,34,38	"	"	"	"
N13	3CLA	"	20	21,5,2,8,24,27,28,12,9, 13,31,37,16,17,34,38	"	"	"	"
N14	3RA	"	16	21,5,2,8,24,27,28,12,9, 13,31,37,20,17,34,38	"	"	"	"

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TABLE XV. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N15	3DB	Input current, high level	17	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,34,38	0	200	nA	14
N16	3CLB	"	34	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,17,38	"	"	"	"
N17	3RB	"	38	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,17,34	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO3 from earth ground.

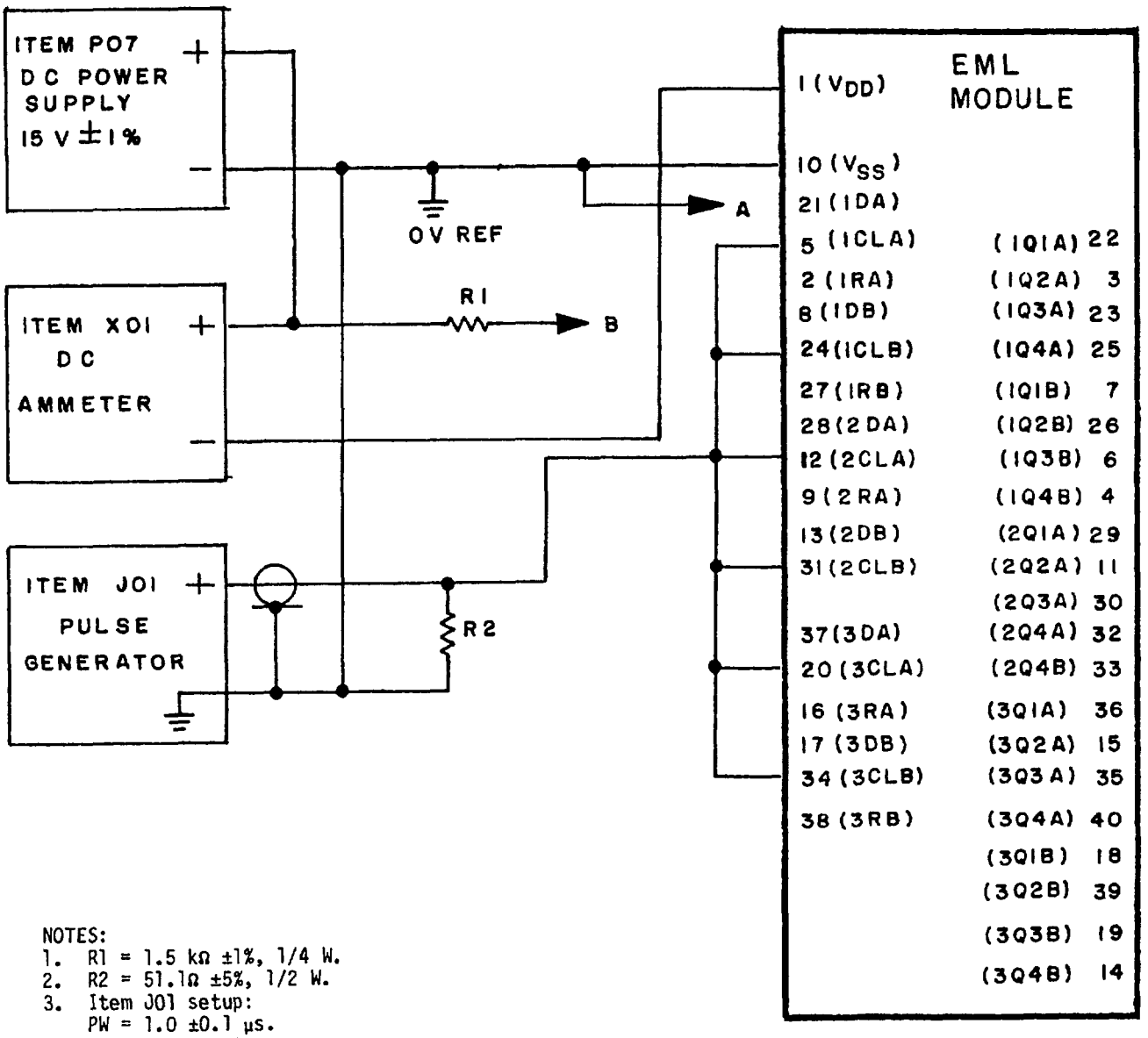
FIGURE 14. Input current (high-level) test circuit for table XV.

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TABLE XVI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply current, low level	21,8,28,13,37,17	Open	0	15	µA	15
P2	Power supply current, high level	Open	21,8,28,13,37,17	0	15	µA	15

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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/2 W.
3. Item JO1 setup:
   PW = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.2 µs.
   PA = 15 V ±1%.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns.
4. Put item JO1 in single pulse mode after all connections are made. Pulse item JO1 four times, measure test condition M1 and repeat procedure for test condition M2.

FIGURE 15. Power supply current (low-level and high-level) test circuit for table XVI.

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TABLE XVII. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1Q1A	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	36	360	ns	16,17
Q2	1Q2A	"	"	"	"	3	"	"	"	"
Q3	1Q3A	"	"	"	"	23	"	"	"	"
Q4	1Q4A	"	"	"	"	25	"	"	"	"
Q5	1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
Q6	1Q2B	"	"	"	"	26	"	"	"	"
Q7	1Q3B	"	"	"	"	6	"	"	"	"
Q8	1Q4B	"	"	"	"	4	"	"	"	"
Q9	2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
Q10	2Q2A	"	"	"	"	11	"	"	"	"
Q11	2Q3A	"	"	"	"	30	"	"	"	"
Q12	2Q4A	"	"	"	"	32	"	"	"	"
Q13	2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
Q14	3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
Q15	3Q2A	"	"	"	"	15	"	"	"	"
Q16	3Q3A	"	"	"	"	35	"	"	"	"
Q17	3Q4A	"	"	"	"	40	"	"	"	"
Q18	3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XVII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q19	3Q2B	Transition time, (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9,13,31,37,20,16,38	34	17	39	36	360	ns	16,17
Q20	3Q3B	"	"	"	"	19	"	"	"	"
Q21	3Q4B	"	"	"	"	14	"	"	"	"
Q22	1Q1A	Transition time, (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31,37,20,16,17,34,38	5	21	22	96	960	"	"
Q23	1Q2A	"	"	"	"	3	"	"	"	"
Q24	1Q3A	"	"	"	"	23	"	"	"	"
Q25	1Q4A	"	"	"	"	25	"	"	"	"
Q26	1Q1B	"	21,5,2,27,28,12,9,13,31,37,20,16,17,34,38	24	8	7	"	"	"	"
Q27	1Q2B	"	"	"	"	26	"	"	"	"
Q28	1Q3B	"	"	"	"	6	"	"	"	"
Q29	1Q4B	"	"	"	"	4	"	"	"	"
Q30	2Q1A	"	21,5,2,8,24,27,9,13,31,37,20,16,17,34,38	12	28	29	"	"	"	"
Q31	2Q2A	"	"	"	"	11	"	"	"	"
Q32	2Q3A	"	"	"	"	30	"	"	"	"
Q33	2Q4A	"	"	"	"	32	"	"	"	"
Q34	2Q4B	"	21,5,2,8,24,27,28,12,9,37,20,16,17,34,38	31	13	33	"	"	"	"
Q35	3Q1A	"	21,5,2,8,24,27,28,12,9,13,31,16,17,34,38	20	37	36	"	"	"	"
Q36	3Q2A	"	"	"	"	15	"	"	"	"
Q37	3Q3A	"	"	"	"	35	"	"	"	"

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TABLE XVII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q38	3Q4A	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	40	96	960	ns	16,17
Q39	3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"
Q40	3Q2B	"	"	"	"	39	"	"	"	"
Q41	3Q3B	"	"	"	"	19	"	"	"	"
Q42	3Q4B	"	"	"	"	14	"	"	"	"

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TABLE XVIII. Transition time (t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q1A	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	20	200	ns	16,17
R2	1Q2A	"	"	"	"	3	"	"	"	"
R3	1Q3A	"	"	"	"	23	"	"	"	"
R4	1Q4A	"	"	"	"	25	"	"	"	"
R5	1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
R6	1Q2B	"	"	"	"	26	"	"	"	"
R7	1Q3B	"	"	"	"	6	"	"	"	"
R8	1Q4B	"	"	"	"	4	"	"	"	"
R9	2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
R10	2Q2A	"	"	"	"	11	"	"	"	"
R11	2Q3A	"	"	"	"	30	"	"	"	"
R12	2Q4A	"	"	"	"	32	"	"	"	"
R13	2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
R14	3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
R15	3Q2A	"	"	"	"	15	"	"	"	"
R16	3Q3A	"	"	"	"	35	"	"	"	"
R17	3Q4A	"	"	"	"	40	"	"	"	"
R18	3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XVIII. Transition time (t_{THL}) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R19	3Q2B	Transition time, (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	39	20	200	ns	16,17
R20	3Q3B	"	"	"	"	19	"	"	"	"
R21	3Q4B	"	"	"	"	14	"	"	"	"
R22	1Q1A	Transition time, (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	50	500	"	"
R23	1Q2A	"	"	"	"	3	"	"	"	"
R24	1Q3A	"	"	"	"	23	"	"	"	"
R25	1Q4A	"	"	"	"	25	"	"	"	"
R26	1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
R27	1Q2B	"	"	"	"	26	"	"	"	"
R28	1Q3B	"	"	"	"	6	"	"	"	"
R29	1Q4B	"	"	"	"	4	"	"	"	"
R30	2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
R31	2Q2A	"	"	"	"	11	"	"	"	"
R32	2Q3A	"	"	"	"	30	"	"	"	"
R33	2Q4A	"	"	"	"	32	"	"	"	"
R34	2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
R35	3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
R36	3Q2A	"	"	"	"	15	"	"	"	"
R37	3Q3A	"	"	"	"	35	"	"	"	"

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TABLE XVIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R38	3Q4A	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	40	50	500	ns	16,17
R39	3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"
R40	3Q2B	"	"	"	"	39	"	"	"	"
R41	3Q3B	"	"	"	"	19	"	"	"	"
R42	3Q4B	"	"	"	"	14	"	"	"	"

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TABLE XIX. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1CLA to 1Q1A	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	75	750	ns	16,18
S2	1CLA to 1Q2A	"	"	"	"	3	"	"	"	"
S3	1CLA to 1Q3A	"	"	"	"	23	"	"	"	"
S4	1CLA to 1Q4A	"	"	"	"	25	"	"	"	"
S5	1CLB to 1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
S6	1CLB to 1Q2B	"	"	"	"	26	"	"	"	"
S7	1CLB to 1Q3B	"	"	"	"	6	"	"	"	"
S8	1CLB to 1Q4B	"	"	"	"	4	"	"	"	"
S9	2CLA to 2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
S10	2CLA to 2Q2A	"	"	"	"	11	"	"	"	"
S11	2CLA to 2Q3A	"	"	"	"	30	"	"	"	"
S12	2CLA to 2Q4A	"	"	"	"	32	"	"	"	"
S13	2CLB to 2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
S14	3CLA to 3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
S15	3CLA to 3Q2A	"	"	"	"	15	"	"	"	"
S16	3CLA to 3Q3A	"	"	"	"	35	"	"	"	"
S17	3CLA to 3Q4A	"	"	"	"	40	"	"	"	"

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Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S18	3CLB to 3Q1B	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	75	750	ns	16,18
S19	3CLB to 3Q2B	"	"	"	"	39	"	"	"	"
S20	3CLB to 3Q3B	"	"	"	"	19	"	"	"	"
S21	3CLB to 3Q4B	"	"	"	"	14	"	"	"	"
S22	1CLA to 1Q1A	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	99	998	"	"
S23	1CLA to 1Q2A	"	"	"	"	3	"	"	"	"
S24	1CLA to 1Q3A	"	"	"	"	23	"	"	"	"
S25	1CLA to 1Q4A	"	"	"	"	25	"	"	"	"
S26	1CLB to 1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
S27	1CLB to 1Q2B	"	"	"	"	26	"	"	"	"
S28	1CLB to 1Q3B	"	"	"	"	6	"	"	"	"
S29	1CLB to 1Q4B	"	"	"	"	4	"	"	"	"
S30	2CLA to 2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
S31	2CLA to 2Q2A	"	"	"	"	11	"	"	"	"
S32	2CLA to 2Q3A	"	"	"	"	30	"	"	"	"
S33	2CLA to 2Q4A	"	"	"	"	32	"	"	"	"
S34	2CLB to 2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
S35	3CLA to 3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"

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TABLE XIX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S36	3CLA to 3Q2A	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	15	99	998	ns	16,18
S37	3CLA to 3Q3A	"	"	"	"	35	"	"	"	"
S38	3CLA to 3Q4A	"	"	"	"	40	"	"	"	"
S39	3CLB to 3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"
S40	3CLB to 3Q2B	"	"	"	"	39	"	"	"	"
S41	3CLB to 3Q3B	"	"	"	"	19	"	"	"	"
S42	3CLB to 3Q4B	"	"	"	"	14	"	"	"	"

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TABLE XX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1CLA to 1Q1A	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	75	750	ns	16,18
T2	1CLA to 1Q2A	"	"	"	"	3	"	"	"	"
T3	1CLA to 1Q3A	"	"	"	"	23	"	"	"	"
T4	1CLA to 1Q4A	"	"	"	"	25	"	"	"	"
T5	1CLB to 1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
T6	1CLA to 1Q2B	"	"	"	"	26	"	"	"	"
T7	1CLA to 1Q3B	"	"	"	"	6	"	"	"	"
T8	1CLA to 1Q4B	"	"	"	"	4	"	"	"	"
T9	2CLA to 2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
T10	2CLA to 2Q2A	"	"	"	"	11	"	"	"	"
T11	2CLA to 2Q3A	"	"	"	"	30	"	"	"	"
T12	2CLA to 2Q4A	"	"	"	"	32	"	"	"	"
T13	2CLB to 2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
T14	3CLA to 3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
T15	3CLA to 3Q2A	"	"	"	"	15	"	"	"	"
T16	3CLA to 3Q3A	"	"	"	"	35	"	"	"	"
T17	3CLA to 3Q4A	"	"	"	"	40	"	"	"	"

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TABLE XX. Propagation delay time (t_{PHL}) test - Continued.

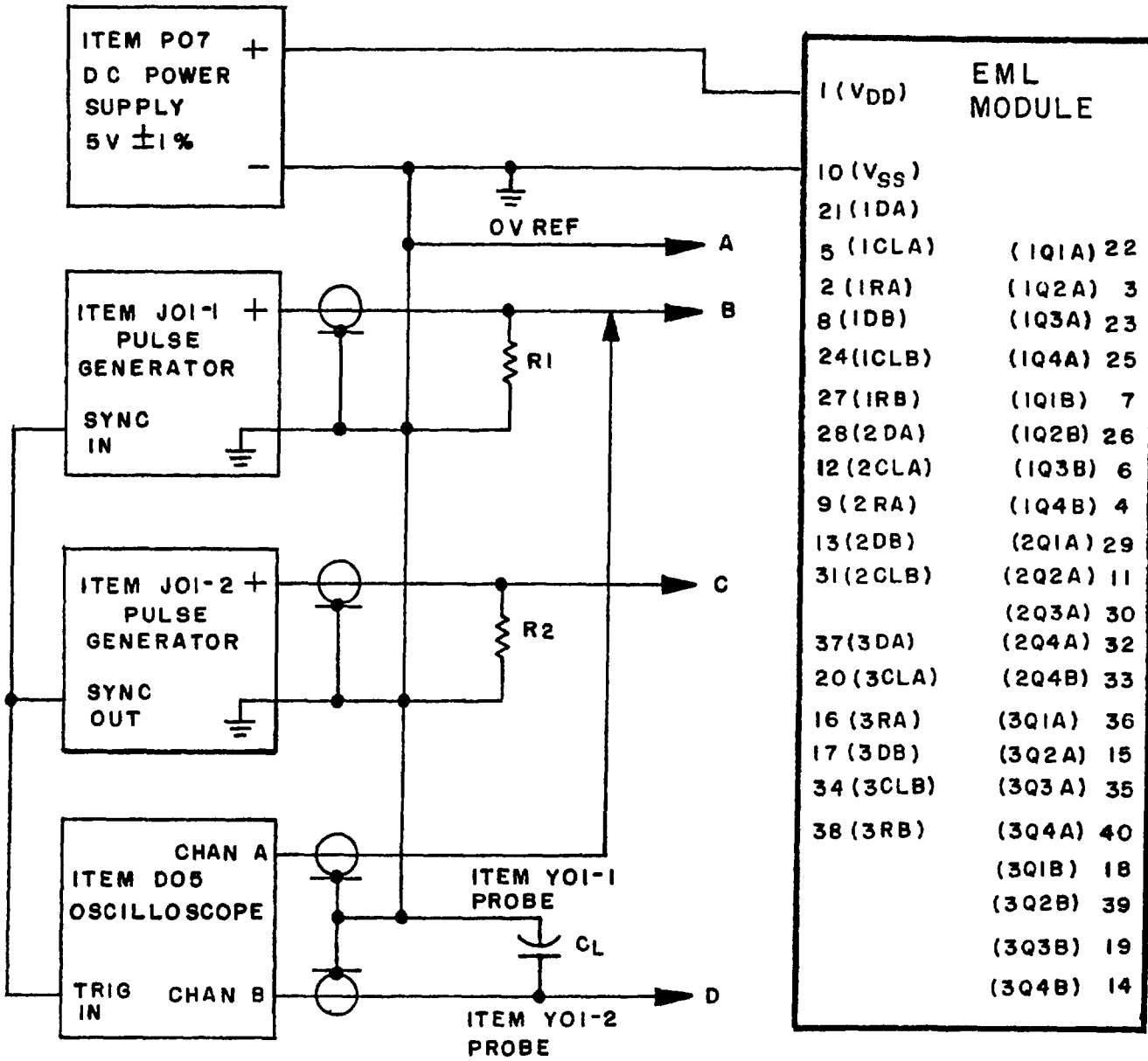
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T18	3CLB to 3Q1B	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	75	750	ns	16,18
T19	3CLB to 3Q2B	"	"	"	"	39	"	"	"	"
T20	3CLB to 3Q3B	"	"	"	"	19	"	"	"	"
T21	3CLB to 3Q4B	"	"	"	"	14	"	"	"	"
T22	1CLA to 1Q1A	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	99	998	"	"
T23	1CLA to 1Q2A	"	"	"	"	3	"	"	"	"
T24	1CLA to 1Q3A	"	"	"	"	23	"	"	"	"
T25	1CLA to 1Q4A	"	"	"	"	25	"	"	"	"
T26	1CLB to 1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
T27	1CLB to 1Q2B	"	"	"	"	26	"	"	"	"
T28	1CLB to 1Q3B	"	"	"	"	6	"	"	"	"
T29	1CLB to 1Q4B	"	"	"	"	4	"	"	"	"
T30	2CLA to 2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
T31	2CLA to 2Q2A	"	"	"	"	11	"	"	"	"
T32	2CLA to 2Q3A	"	"	"	"	30	"	"	"	"
T33	2CLA to 2Q4A	"	"	"	"	32	"	"	"	"
T34	2CLB to 2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
T35	3CLA to 3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"

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TABLE XX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T36	3CLA to 3Q2A Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	15	99	998	ns	16,18
T37	3CLA to 3Q3A "	"	"	"	35	"	"	"	"
T38	3CLA to 3Q4A "	"	"	"	40	"	"	"	"
T39	3CLB to 3Q1B "	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"
T40	3CLB to 3Q2B "	"	"	"	39	"	"	"	"
T41	3CLB to 3Q3B "	"	"	"	19	"	"	"	"
T42	3CLB to 3Q4B "	"	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions Q1-Q21, R1-R21, S1-S21, and T1-T21.
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions Q22-Q42, R22-R42, S22-S42, and T22-T42.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVII through XX.

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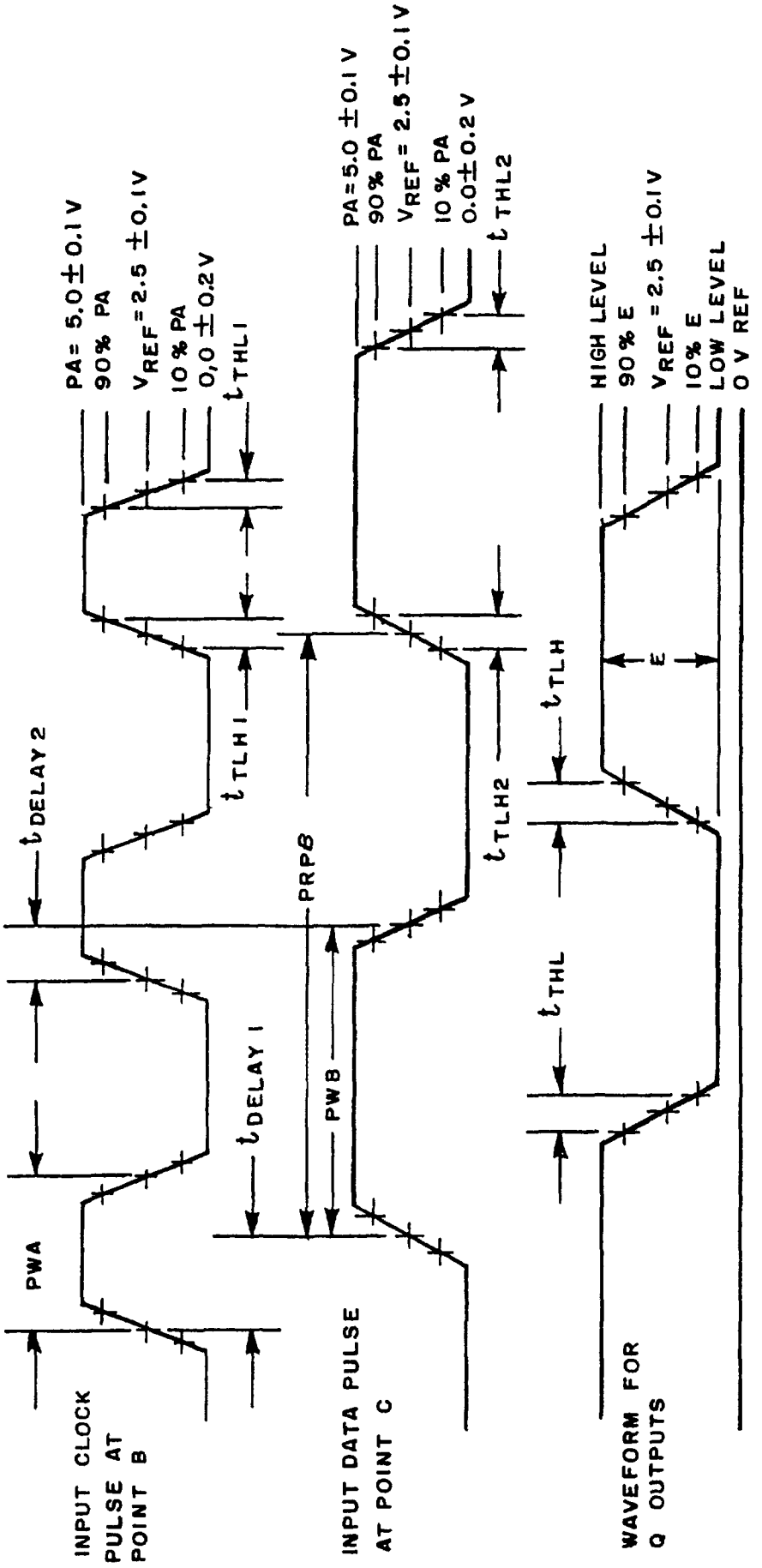


FIGURE 17. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XVII and XVIII.

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NOTES:
1. Clock input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in external trigger mode.
2. Data input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 500 +50, -0 ns.
4. Before making measurements, verify that the output pulse high level is ≥ 4.6 V and that the low level is ≤ 0.4 V.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 17. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XVII and XVIII - Continued.

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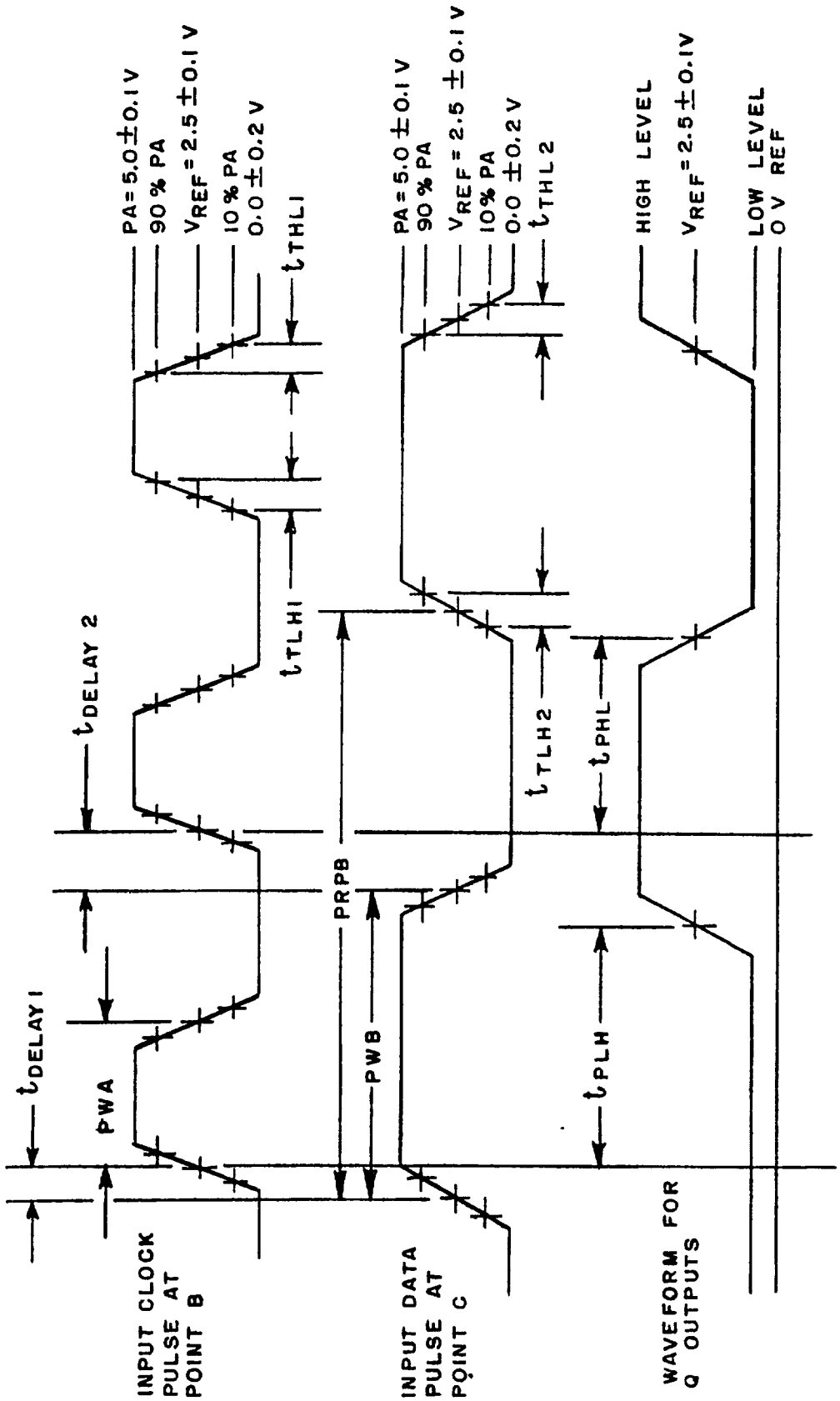


FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIX and XX.

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NOTES:
1. Clock input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in double pulse mode.
2. Data input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 500 +50, -0 ns.
4. Before making measurements, verify that the output pulse high level is $\geq$ 4.6 V and that the low level is $\leq$ 0.4 V.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIX and XX - Continued.

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TABLE XXI. Propagation delay time (t_{PHL}) test.

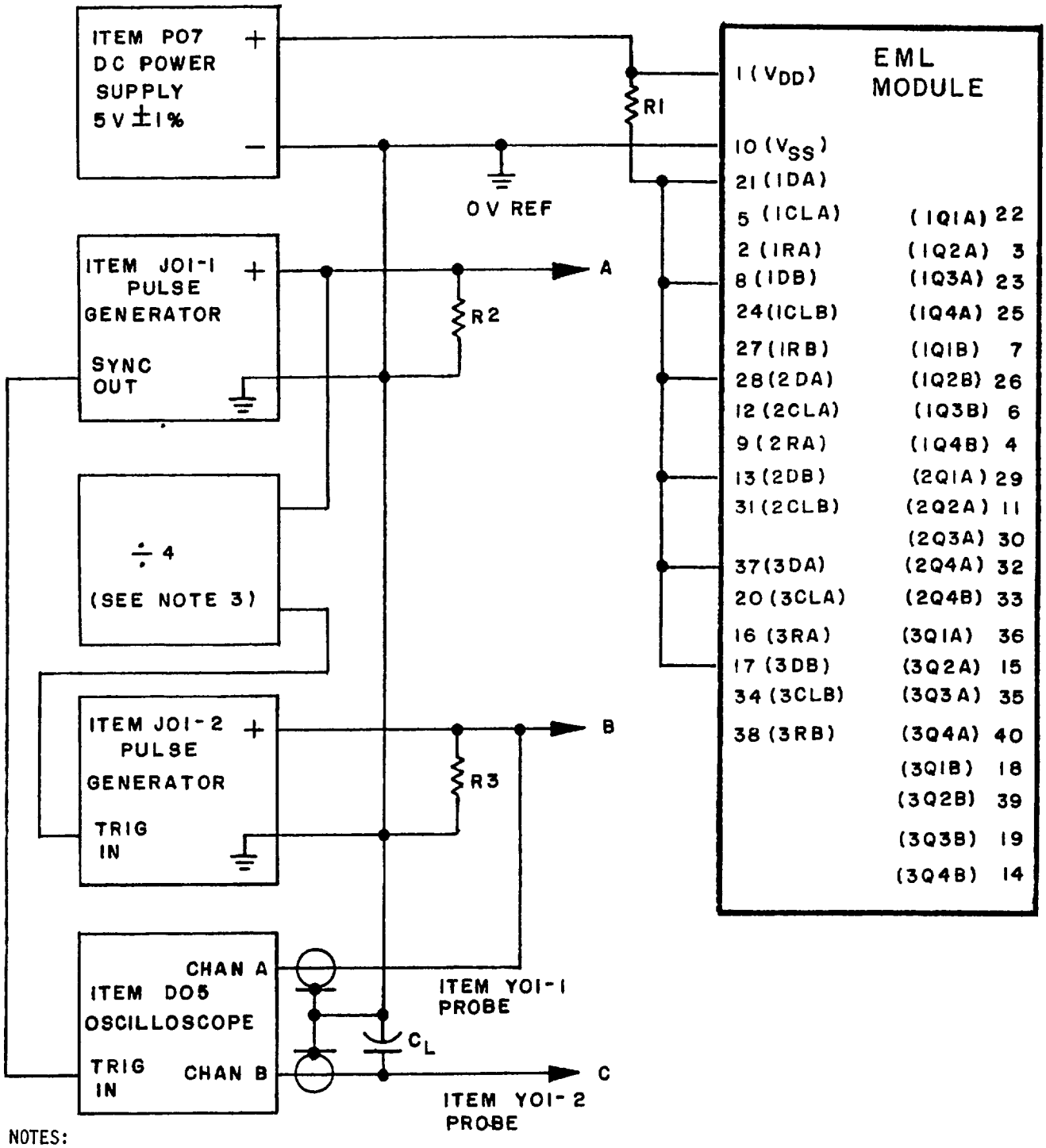
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	1RA to 1Q1A Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	5,24,27,12,9,31, 20,16,34,38	2	22	75	750	ns	19,20
U2	1RA to 1Q2A "	"	"	3	"	"	"	"
U3	1RA to 1Q3A "	"	"	23	"	"	"	"
U4	1RA to 1Q4A "	"	"	25	"	"	"	"
U5	1RB to 1Q1B "	24,5,2,12,9,31, 20,16,34,38	27	7	"	"	"	"
U6	1RB to 1Q2B "	"	"	26	"	"	"	"
U7	1RB to 1Q3B "	"	"	6	"	"	"	"
U8	1RB to 1Q4B "	"	"	4	"	"	"	"
U9	2RA to 2Q1A "	12,5,2,24,27,31, 20,16,34,38	9	29	"	"	"	"
U10	2RA to 2Q2A "	"	"	11	"	"	"	"
U11	2RA to 2Q3A "	"	"	30	"	"	"	"
U12	2RA to 2Q4A "	"	"	32	"	"	"	"
U13	3RA to 3Q1A "	5,2,24,27,12,9, 31,34,38	16	36	"	"	"	"
U14	3RA to 3Q2A "	"	"	15	"	"	"	"
U15	3RA to 3Q3A "	"	"	35	"	"	"	"
U16	3RA to 3Q4A "	"	"	40	"	"	"	"
U17	3RB to 3Q1B "	5,2,24,27,12,9, 31,20,16	38	18	"	"	"	"
U18	3RB to 3Q2B "	"	"	39	"	"	"	"
U19	3RB to 3Q3B "	"	"	19	"	"	"	"
U20	3RB to 3Q4B "	"	"	14	"	"	"	"

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
U21	1RA to 1Q1A	Propagation delay time, (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	5,24,27,12,9,31, 20,16,34,38	2	22	99	998	ns	19,20
U22	1RA to 1Q2A	"	"	"	3	"	"	"	"
U23	1RA to 1Q3A	"	"	"	23	"	"	"	"
U24	1RA to 1Q4A	"	"	"	25	"	"	"	"
U25	1RB to 1Q1B	"	24,5,2,12,9,31, 20,16,34,38	27	7	"	"	"	"
U26	1RB to 1Q2B	"	"	"	26	"	"	"	"
U27	1RB to 1Q3B	"	"	"	6	"	"	"	"
U28	1RB to 1Q4B	"	"	"	4	"	"	"	"
U29	2RA to 2Q1A	"	12,5,2,24,27,31, 20,16,34,38	9	29	"	"	"	"
U30	2RA to 2Q2A	"	"	"	11	"	"	"	"
U31	2RA to 2Q3A	"	"	"	30	"	"	"	"
U32	2RA to 2Q4A	"	"	"	32	"	"	"	"
U33	3RA to 3Q1A	"	5,2,24,27,12,9,20, 31,34,38	16	36	"	"	"	"
U34	3RA to 3Q2A	"	"	"	15	"	"	"	"
U35	3RA to 3Q3A	"	"	"	35	"	"	"	"
U36	3RA to 3Q4A	"	"	"	40	"	"	"	"
U37	3RB to 3Q1B	"	5,2,24,27,12,9, 31,20,16,34	38	18	"	"	"	"
U38	3RB to 3Q2B	"	"	"	39	"	"	"	"
U39	3RB to 3Q3B	"	"	"	19	"	"	"	"
U40	3RB to 3Q4B	"	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF for test conditions U1-U20.
   = 150 ±15 pF for test conditions U21-U40.
3. The divide by 4 function may be either a commercially available unit (such as the Tektronix Model 00501) or any circuit that will provide one trigger input to item J01-2 for every fourth output pulse of item J01-1.

FIGURE 19. Propagation delay time (t_{PHL}) test circuit for table XXI.

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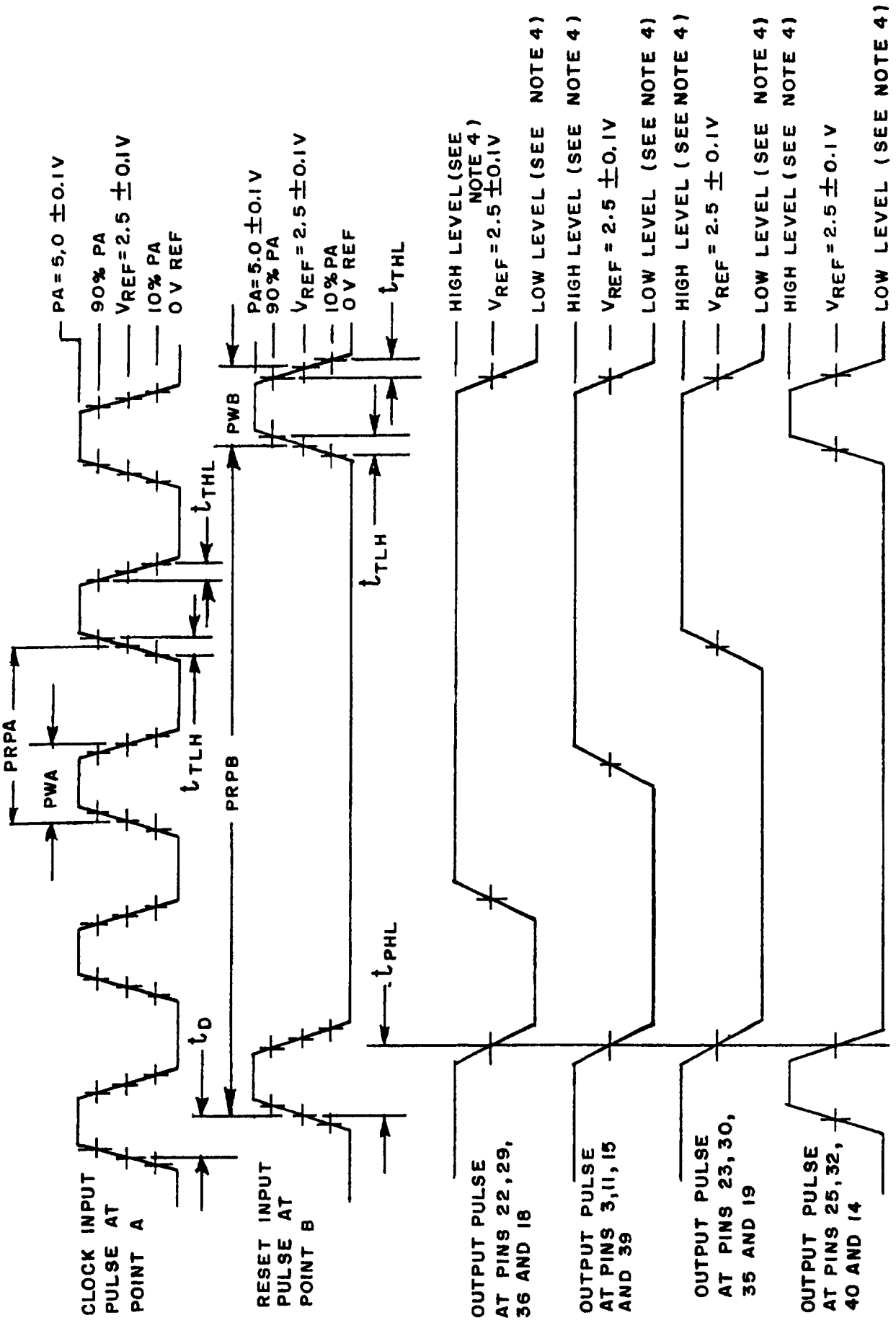


FIGURE 20. Propagation delay time (t_{PHL}) test circuit waveforms for table XXI.

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NOTES:
1. Item J01-1 input pulse parameters:
   PWA = 500 ±50 ns.
   PRPA = 2,000 ±200 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Item J01-2 input pulse parameters:
   PWB = 400 +8 ns, -0 ns.
   PRPB = 8,000 ±800 ns.
   $t_D = 500 \pm 50$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
3. Before making measurements specified, verify that the output pulse high level is $\geq$ 4.6 V and the low level is $\leq$ 0.4 V.
4. Insure that the output pulse is $\geq$ 4.5 V for high level and $\leq$ 0.5 V for low level.

FIGURE 20. Propagation delay time (t_{PHL}) test circuit waveforms for table XXI - Continued.

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TABLE XXII. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1Q1A	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	18	180	ns	21,22
V2	1Q2A	"	"	"	"	3	"	"	"	"
V3	1Q3A	"	"	"	"	23	"	"	"	"
V4	1Q4A	"	"	"	"	25	"	"	"	"
V5	1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
V6	1Q2B	"	"	"	"	26	"	"	"	"
V7	1Q3B	"	"	"	"	6	"	"	"	"
V8	1Q4B	"	"	"	"	4	"	"	"	"
V9	2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
V10	2Q2A	"	"	"	"	11	"	"	"	"
V11	2Q3A	"	"	"	"	30	"	"	"	"
V12	2Q4A	"	"	"	"	32	"	"	"	"
V13	2Q4B	"	21,5,2,8,24,27,28,12,9,37, 20,16,17,34,38	31	13	33	"	"	"	"
V14	3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
V15	3Q2A	"	"	"	"	15	"	"	"	"
V16	3Q3A	"	"	"	"	35	"	"	"	"
V17	3Q4A	"	"	"	"	40	"	"	"	"
V18	3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V19	3Q2B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	39	18	180	ns	21,22
V20	3Q3B "	"	"	"	19	"	"	"	"
V21	3Q4B "	"	"	"	14	"	"	"	"

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TABLE XXIII. Transition time (t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1Q1A	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	10	100	ns	21,22
W2	1Q2A	"	"	"	"	3	"	"	"	"
W3	1Q3A	"	"	"	"	23	"	"	"	"
W4	1Q4A	"	"	"	"	25	"	"	"	"
W5	1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
W6	1Q2B	"	"	"	"	26	"	"	"	"
W7	1Q3B	"	"	"	"	6	"	"	"	"
W8	1Q4B	"	"	"	"	4	"	"	"	"
W9	2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
W10	2Q2A	"	"	"	"	11	"	"	"	"
W11	2Q3A	"	"	"	"	30	"	"	"	"
W12	2Q4A	"	"	"	"	32	"	"	"	"
W13	2Q4B	"	21,5,2,8,24,27,28,12,9,37, 20,16,17,34,38	31	13	33	"	"	"	"
W14	3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
W15	3Q2A	"	"	"	"	15	"	"	"	"
W16	3Q3A	"	"	"	"	35	"	"	"	"
W17	3Q4A	"	"	"	"	40	"	"	"	"
W18	3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W19	3Q2B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	39	10	100	ns	21,22
W20	3Q3B "	"	"	"	19	"	"	"	"
W21	3Q4B "	"	"	"	14	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1CLA to 1Q1A Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	37	375	ns	21,23
X2	1CLA to 1Q2A "	"	"	"	3	"	"	"	"
X3	1CLA to 1Q3A "	"	"	"	23	"	"	"	"
X4	1CLA to 1Q4A "	"	"	"	25	"	"	"	"
X5	1CLB to 1Q1B "	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
X6	1CLB to 1Q2B "	"	"	"	26	"	"	"	"
X7	1CLB to 1Q3B "	"	"	"	6	"	"	"	"
X8	1CLB to 1Q4B "	"	"	"	4	"	"	"	"
X9	2CLA to 2Q1A "	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
X10	2CLA to 2Q2A "	"	"	"	11	"	"	"	"
X11	2CLA to 2Q3A "	"	"	"	30	"	"	"	"
X12	2CLA to 2Q4A "	"	"	"	32	"	"	"	"
X13	2CLB to 2Q4B "	21,5,2,8,24,27,28,12,9,37, 20,16,17,34,38	31	13	33	"	"	"	"
X14	3CLA to 3Q1A "	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
X15	3CLA to 3Q2A "	"	"	"	15	"	"	"	"
X16	3CLA to 3Q3A "	"	"	"	35	"	"	"	"
X17	3CLA to 3Q4A "	"	"	"	40	"	"	"	"
X18	3CLB to 3Q1B "	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X19	3CLB to 3Q2B Propagation delay time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	39	37	375	ns	21,23
X20	3CLB to 3Q3B "	"	"	"	19	"	"	"	"
X21	3CLB to 3Q4B "	"	"	"	14	"	"	"	"

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TABLE XXV. Propagation delay time (t_{PHL}) test.

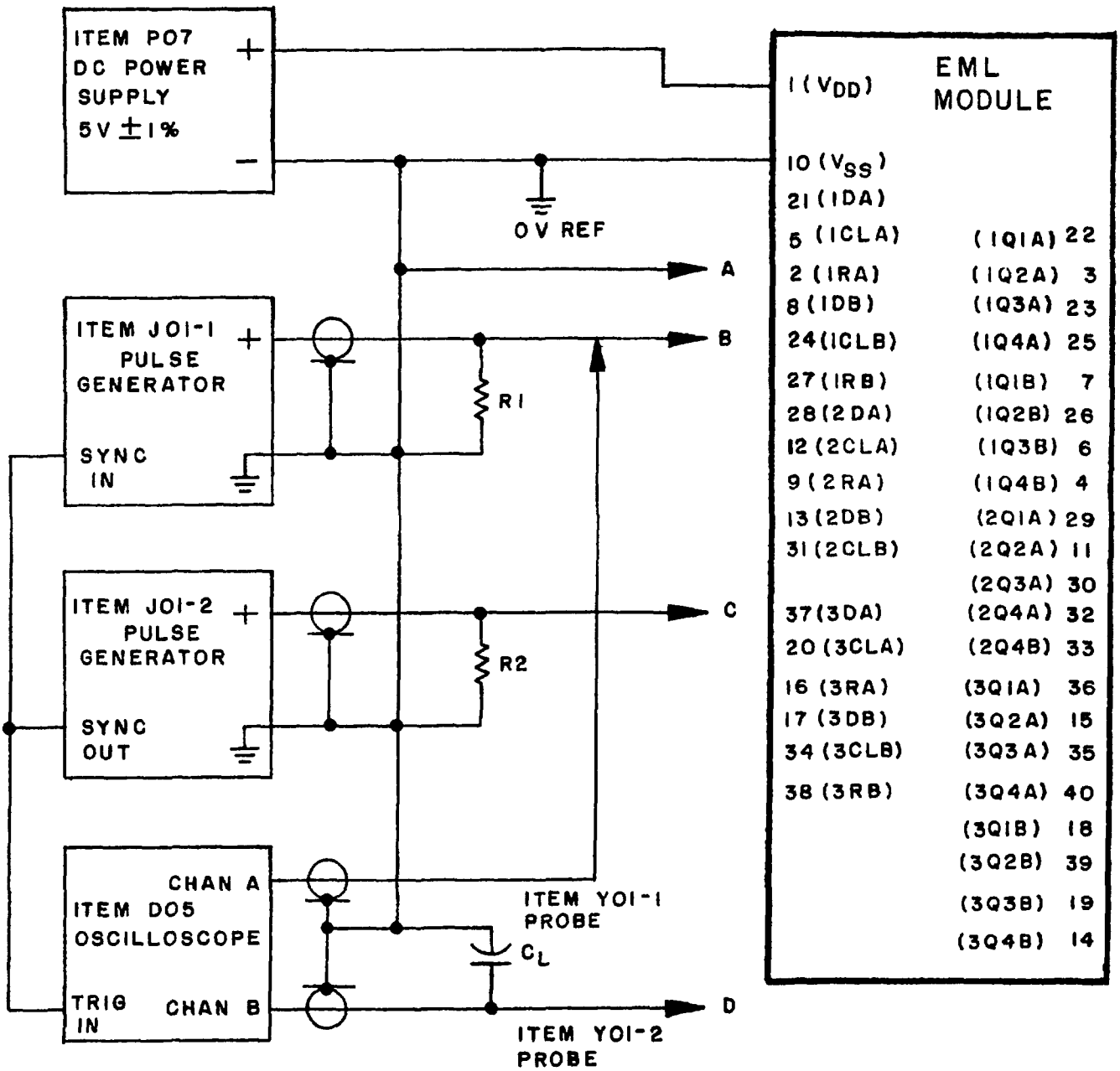
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	1CLA to 1Q1A Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	2,8,24,27,28,12,9,13,31, 37,20,16,17,34,38	5	21	22	37	375	ns	21,23
Y2	1CLA to 1Q2A "	"	"	"	3	"	"	"	"
Y3	1CLA to 1Q3A "	"	"	"	23	"	"	"	"
Y4	1CLA to 1Q4A "	"	"	"	25	"	"	"	"
Y5	1CLB to 1Q1B "	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
Y6	1CLB to 1Q2B "	"	"	"	26	"	"	"	"
Y7	1CLB to 1Q3B "	"	"	"	6	"	"	"	"
Y8	1CLB to 1Q4B "	"	"	"	4	"	"	"	"
Y9	2CLA to 2Q1A "	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
Y10	2CLA to 2Q2A "	"	"	"	11	"	"	"	"
Y11	2CLA to 2Q3A "	"	"	"	30	"	"	"	"
Y12	2CLA to 2Q4A "	"	"	"	32	"	"	"	"
Y13	2CLB to 2Q4B "	21,5,2,8,24,27,28,12,9,37, 20,16,17,34,38	31	13	33	"	"	"	"
Y14	3CLA to 3Q1A "	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
Y15	3CLA to 3Q2A "	"	"	"	15	"	"	"	"
Y16	3CLA to 3Q3A "	"	"	"	35	"	"	"	"
Y17	3CLA to 3Q4A "	"	"	"	40	"	"	"	"
Y18	3CLB to 3Q1B "	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y19	3CLB to 3Q2B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	39	37	375	ns	21,23
Y20	3CLB to 3Q3B "	"	"	"	19	"	"	"	"
Y21	3CLB to 3Q4B "	"	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXII through XXV.

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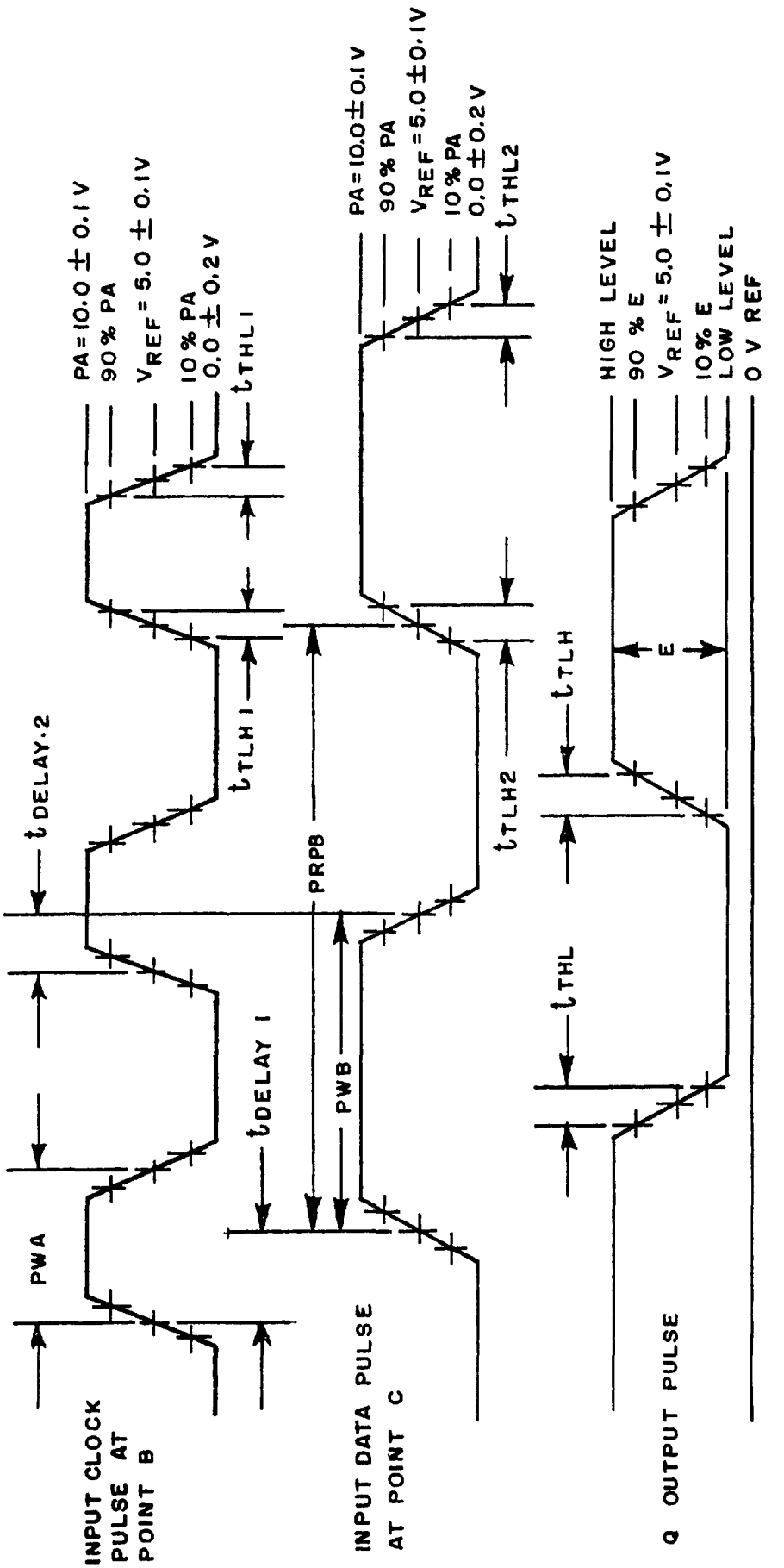


FIGURE 22. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXII and XXIII.

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NOTES:
1. Clock input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in external trigger mode.
2. Data input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 500 +50, -0 ns.
4. Before making measurements, verify that the output pulse high level ≥ 9 V and that the low level is ≤ 1 V.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXII and XXIII - Continued.

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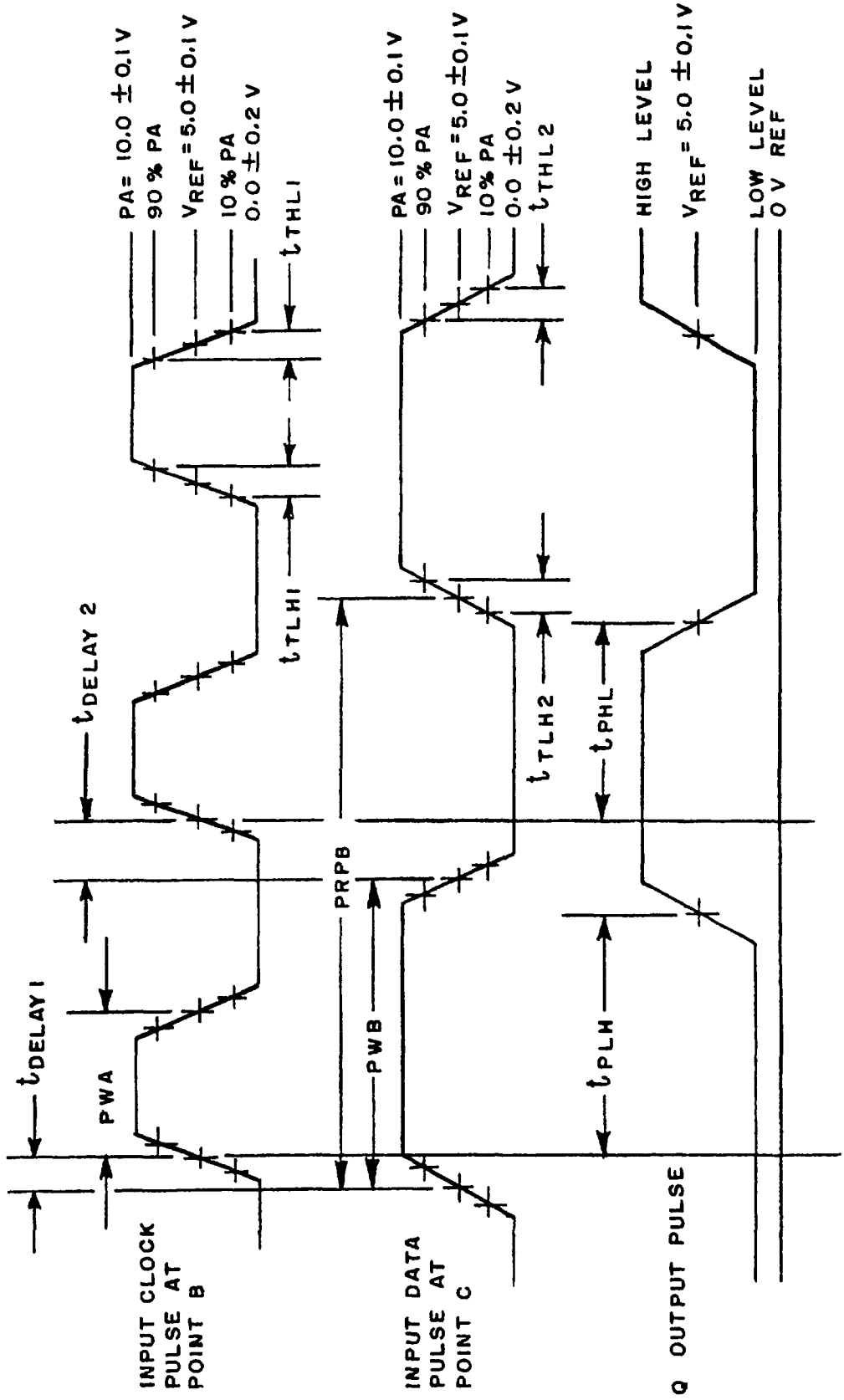


FIGURE 23. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIV and XXV.

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NOTES:
1. Clock input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in external trigger mode.
2. Data input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 500 +50, -0 ns.
4. Before making measurements, verify that the output pulse high level ≥ 9 V and that the low level is ≤ 1 V.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

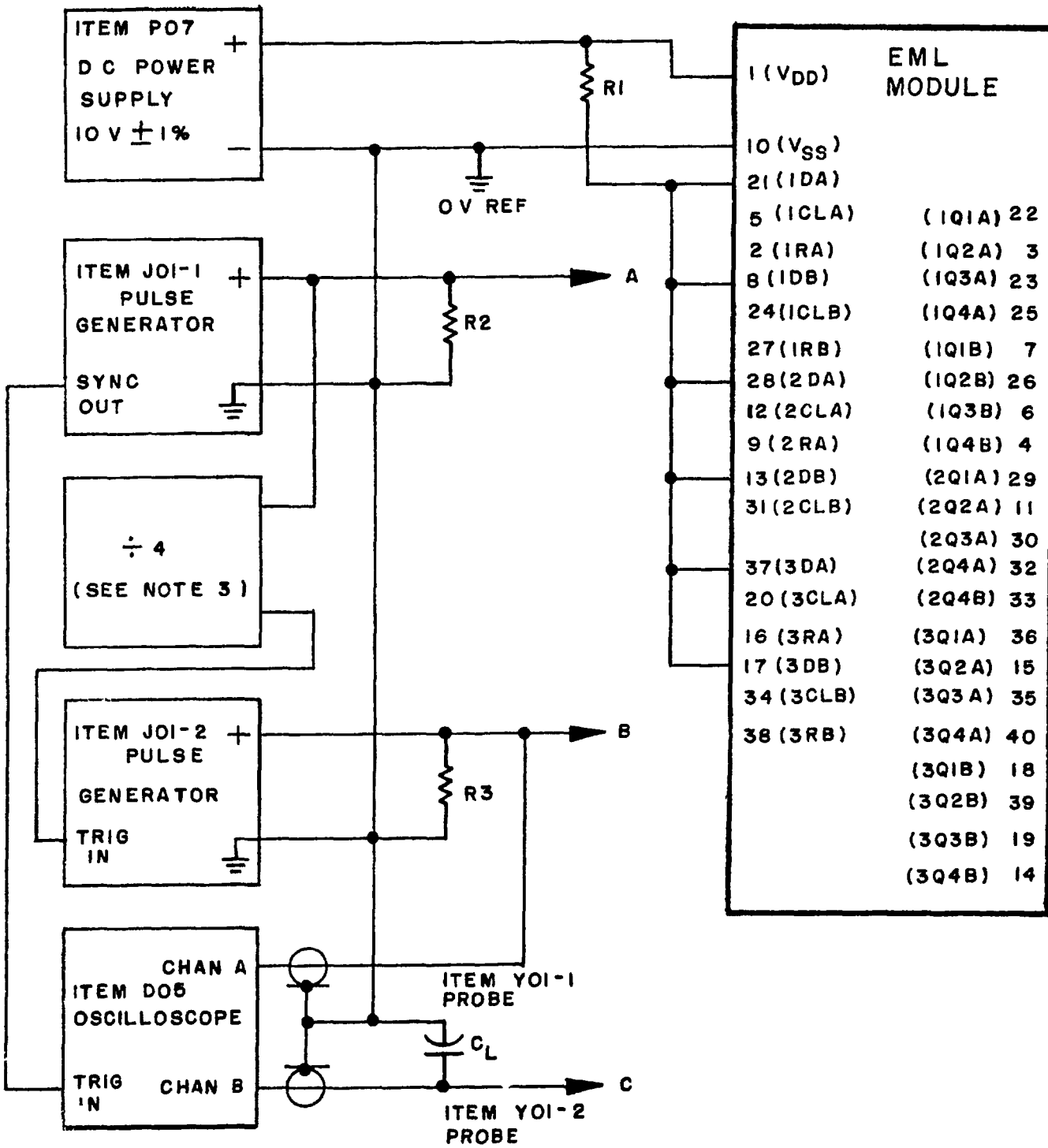
FIGURE 23. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIV and XXV - Continued.

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TABLE XXVI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	1RA to 1Q1A Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	5,24,27,12,9,31,20,16 34,38	2	22	37	375	ns	24,25
Z2	1RA to 1Q2A "	"	"	3	"	"	"	"
Z3	1RA to 1Q3A "	"	"	23	"	"	"	"
Z4	1RA to 1Q4A "	"	"	25	"	"	"	"
Z5	1RB to 1Q1B "	24,5,2,12,9,31,20,16, 34,38	27	7	"	"	"	"
Z6	1RB to 1Q2B "	"	"	26	"	"	"	"
Z7	1RB to 1Q3B "	"	"	6	"	"	"	"
Z8	1RB to 1Q4B "	"	"	4	"	"	"	"
Z9	2RA to 2Q1A "	12,5,2,24,27,31,20, 16,34,38	9	29	"	"	"	"
Z10	2RA to 2Q2A "	"	"	11	"	"	"	"
Z11	2RA to 2Q3A "	"	"	30	"	"	"	"
Z12	2RA to 2Q4A "	"	"	32	"	"	"	"
Z13	3RA to 3Q1A "	5,2,24,27,12,9,31, 34,38	16	36	"	"	"	"
Z14	3RA to 3Q2A "	"	"	15	"	"	"	"
Z15	3RA to 3Q3A "	"	"	35	"	"	"	"
Z16	3RA to 3Q4A "	"	"	40	"	"	"	"
Z17	3RB to 3Q1B "	5,2,24,27,12,9,31, 20,16	38	18	"	"	"	"
Z18	3RB to 3Q2B "	"	"	39	"	"	"	"
Z19	3RB to 3Q3B "	"	"	19	"	"	"	"
Z20	3RB to 3Q4B "	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).
3. The divide by 4 function may be either a commercially available unit (such as the Tektronix Model 00501) or any circuit that will provide one trigger input to item J01-2 for every fourth output pulse of item J01-1.

FIGURE 24. Propagation delay time (t_{PHL}) test circuit for table XXVI.

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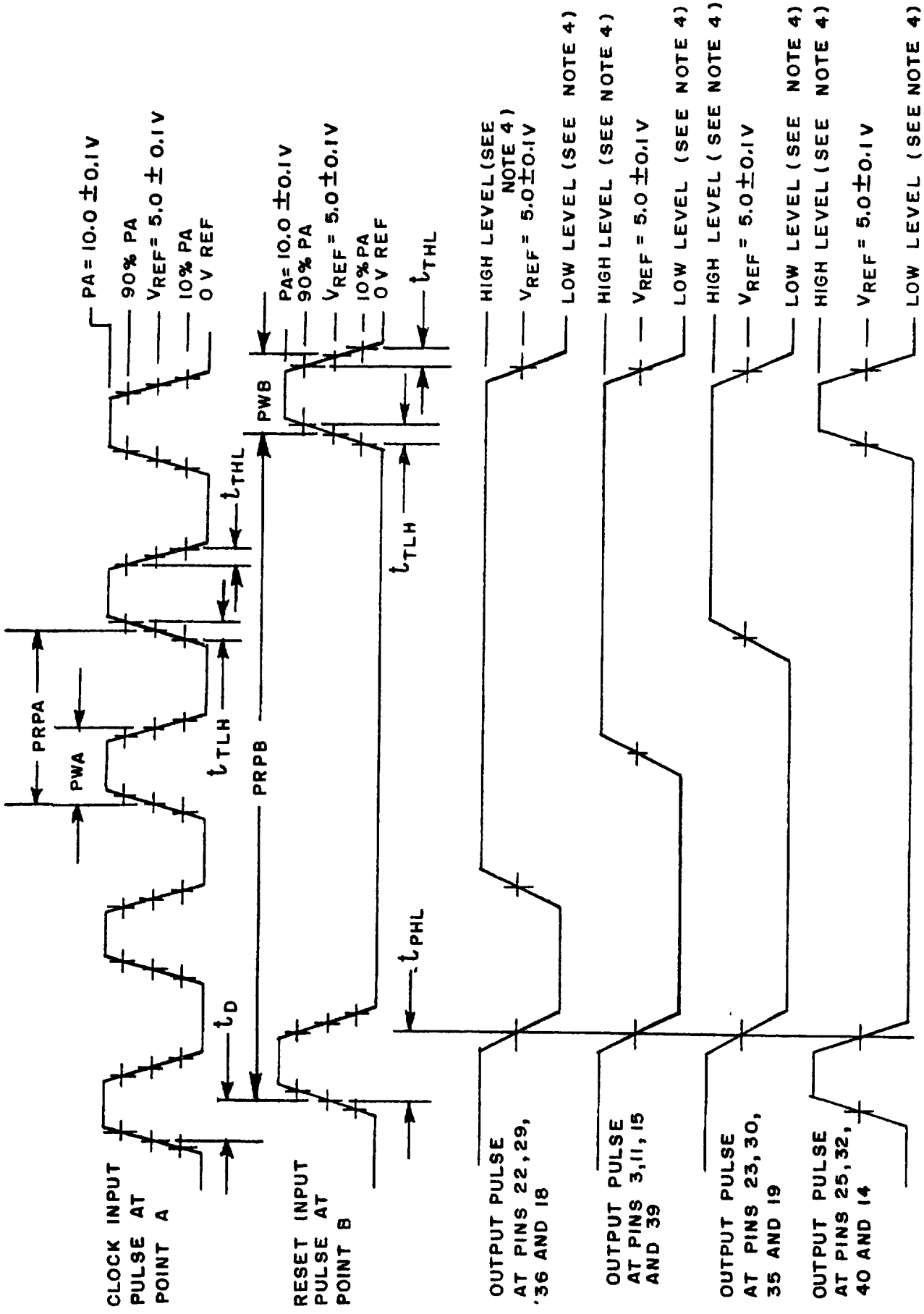


FIGURE 25. Propagation delay time (t_{PHL}) test circuit waveforms for table XXVI.

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NOTES:
1. Item J01-1 input pulse parameters:
   PWA = 500 ±50 ns.
   PRPA = 2,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWB = 400 +8 ns, -0 ns.
   PRPB = 8,000 ±800 ns.
   t_D = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Before making measurements specified, verify that the output pulse high level is ≥ 9.5 V and the low level is ≤ 0.5 V.
4. Insure that the output pulse is ≥ 9 V for high level and ≤ 1 V for low level.

FIGURE 25. Propagation delay time (t_{PHL}) test circuit waveforms for table XXVI - Continued.

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TABLE XXVII. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	1Q1A	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	2,8,24,27,28,12,9,13,31,37,20,16,17,34,38	5	21	22	13	130	ns	26,27
AA2	1Q2A	"	"	"	"	3	"	"	"	"
AA3	1Q3A	"	"	"	"	23	"	"	"	"
AA4	1Q4A	"	"	"	"	25	"	"	"	"
AA5	1Q1B	"	21,5,2,27,28,12,9,13,31,37,20,16,17,34,38	24	8	7	"	"	"	"
AA6	1Q2B	"	"	"	"	26	"	"	"	"
AA7	1Q3B	"	"	"	"	6	"	"	"	"
AA8	1Q4B	"	"	"	"	4	"	"	"	"
AA9	2Q1A	"	21,5,2,8,24,27,9,13,31,37,20,16,17,34,38	12	28	29	"	"	"	"
AA10	2Q2A	"	"	"	"	11	"	"	"	"
AA11	2Q3A	"	"	"	"	30	"	"	"	"
AA12	2Q4A	"	"	"	"	32	"	"	"	"
AA13	2Q4B	"	21,5,2,8,24,27,28,12,9,37,20,16,17,34,38	31	13	33	"	"	"	"
AA14	3Q1A	"	21,5,2,8,24,27,28,12,9,13,31,16,17,34,38	20	37	36	"	"	"	"
AA15	3Q2A	"	"	"	"	15	"	"	"	"
AA16	3Q3A	"	"	"	"	35	"	"	"	"
AA17	3Q4A	"	"	"	"	40	"	"	"	"
AA18	3Q1B	"	21,5,2,8,24,27,28,12,9,13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXVII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
AA19	3Q2B	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	21,5,2,8,24,27,28,12,9,13, 31,37,20,16,38	34	17	39	13	130	ns	26,27
AA20	3Q3B	"	"	"	"	19	"	"	"	"
AA21	3Q4B	"	"	"	"	14	"	"	"	"

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TABLE XXVIII. Transition time (t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
BB1	1Q1A	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	2,8,24,27,28,12,9,13,31,37,20,16,17,34,38	5	21	22	8	80	ns	26,27
BB2	1Q2A	"	"	"	"	3	"	"	"	"
BB3	1Q3A	"	"	"	"	23	"	"	"	"
BB4	1Q4A	"	"	"	"	25	"	"	"	"
BB5	1Q1B	"	21,5,2,27,28,12,9,13,31,37,20,16,17,34,38	24	8	7	"	"	"	"
BB6	1Q2B	"	"	"	"	26	"	"	"	"
BB7	1Q3B	"	"	"	"	6	"	"	"	"
BB8	1Q4B	"	"	"	"	4	"	"	"	"
BB9	2Q1A	"	21,5,2,8,24,27,9,13,31,37,20,16,17,34,38	12	28	29	"	"	"	"
BB10	2Q2A	"	"	"	"	11	"	"	"	"
BB11	2Q3A	"	"	"	"	30	"	"	"	"
BB12	2Q4A	"	"	"	"	32	"	"	"	"
BB13	2Q4B	"	21,5,2,8,24,27,28,12,9,37,20,16,17,34,38	31	13	33	"	"	"	"
BB14	3Q1A	"	21,5,2,8,24,27,28,12,9,13,31,16,17,34,38	20	37	36	"	"	"	"
BB15	3Q2A	"	"	"	"	15	"	"	"	"
BB16	3Q3A	"	"	"	"	35	"	"	"	"
BB17	3Q4A	"	"	"	"	40	"	"	"	"
BB18	3Q1B	"	21,5,2,8,24,27,28,12,9,13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXVIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
BB19	3Q2B	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	21,5,2,8,24,27,28,12,9,13, 31,37,20,16,38	34	17	39	8	80	ns	26,27
BB20	3Q3B	"	"	"	"	19	"	"	"	"
BB21	3Q4B	"	"	"	"	14	"	"	"	"

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TABLE XXIX. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
CC1	1CLA to 1Q1A Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	2,8,24,27,28,12,9,13,31,37,20, 16,17,34,38	5	21	22	17	170	ns	26,28
CC2	1CLA to 1Q2A "	"	"	"	3	"	"	"	"
CC3	1CLA to 1Q3A "	"	"	"	23	"	"	"	"
CC4	1CLA to 1Q4A "	"	"	"	25	"	"	"	"
CC5	1CLB to 1Q1B "	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
CC6	1CLB to 1Q2B "	"	"	"	26	"	"	"	"
CC7	1CLB to 1Q3B "	"	"	"	6	"	"	"	"
CC8	1CLB to 1Q4B "	"	"	"	4	"	"	"	"
CC9	2CLA to 2Q1A "	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
CC10	2CLA to 2Q2A "	"	"	"	11	"	"	"	"
CC11	2CLA to 2Q3A "	"	"	"	30	"	"	"	"
CC12	2CLA to 2Q4A "	"	"	"	32	"	"	"	"
CC13	2CLB to 2Q4B "	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
CC14	3CLA to 3Q1A "	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
CC15	3CLA to 3Q2A "	"	"	"	15	"	"	"	"
CC16	3CLA to 3Q3A "	"	"	"	35	"	"	"	"
CC17	3CLA to 3Q4A "	"	"	"	40	"	"	"	"
CC18	3CLB to 3Q1B "	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXIX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
CC19	3CLB to 3Q2B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,5,2,8,24,27,28,12,9,13, 31,37,20,16,38	34	17	39	17	170	ns	26,28
CC20	3CLB to 3Q3B "	"	"	"	19	"	"	"	"
CC21	3CLB to 3Q4B "	"	"	"	14	"	"	"	"

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TABLE XXX. Propagation delay time (t_{PHL}) test.

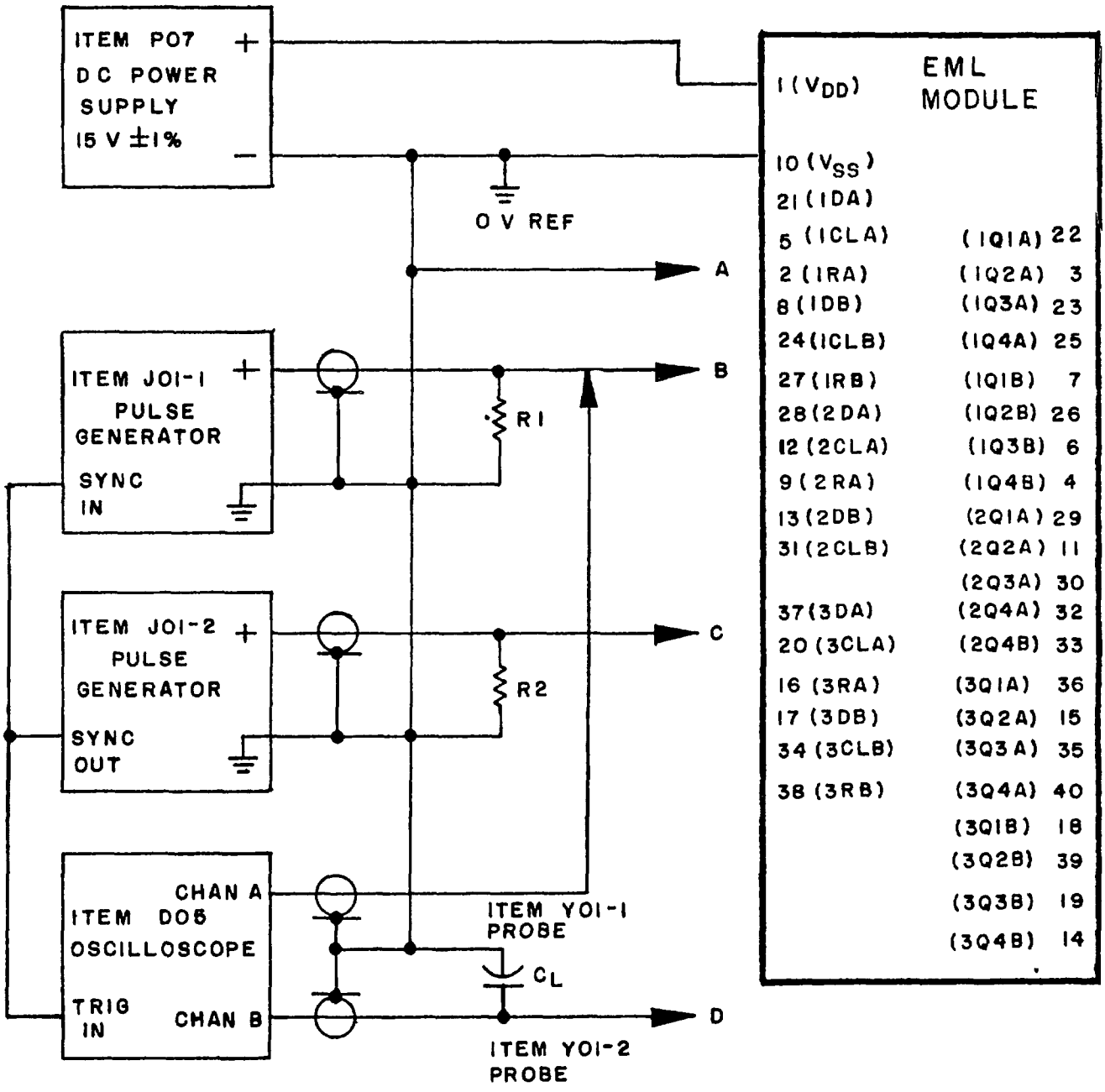
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
DD1	1CLA to 1Q1A	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	2,8,24,27,28,12,9,13,31,37,20, 16,17,34,38	5	21	22	17	170	ns	26,28
DD2	1CLA to 1Q2A	"	"	"	"	3	"	"	"	"
DD3	1CLA to 1Q3A	"	"	"	"	23	"	"	"	"
DD4	1CLA to 1Q4A	"	"	"	"	25	"	"	"	"
DD5	1CLB to 1Q1B	"	21,5,2,27,28,12,9,13,31, 37,20,16,17,34,38	24	8	7	"	"	"	"
DD6	1CLB to 1Q2B	"	"	"	"	26	"	"	"	"
DD7	1CLB to 1Q3B	"	"	"	"	6	"	"	"	"
DD8	1CLB to 1Q4B	"	"	"	"	4	"	"	"	"
DD9	2CLA to 2Q1A	"	21,5,2,8,24,27,9,13,31, 37,20,16,17,34,38	12	28	29	"	"	"	"
DD10	2CLA to 2Q2A	"	"	"	"	11	"	"	"	"
DD11	2CLA to 2Q3A	"	"	"	"	30	"	"	"	"
DD12	2CLA to 2Q4A	"	"	"	"	32	"	"	"	"
DD13	2CLB to 2Q4B	"	21,5,2,8,24,27,28,12,9, 37,20,16,17,34,38	31	13	33	"	"	"	"
DD14	3CLA to 3Q1A	"	21,5,2,8,24,27,28,12,9, 13,31,16,17,34,38	20	37	36	"	"	"	"
DD15	3CLA to 3Q2A	"	"	"	"	15	"	"	"	"
DD16	3CLA to 3Q3A	"	"	"	"	35	"	"	"	"
DD17	3CLA to 3Q4A	"	"	"	"	40	"	"	"	"
DD18	3CLB to 3Q1B	"	21,5,2,8,24,27,28,12,9, 13,31,37,20,16,38	34	17	18	"	"	"	"

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TABLE XXX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
DD19	3CLB to 3Q2B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,5,2,8,24,27,28,12,9,13, 31,37,20,16,38	34	17	39	17	170	ns	26,28
DD20	3CLB to 3Q3B "	"	"	"	19	"	"	"	"
DD21	3CLB to 3Q4B "	"	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 26. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXVII through XXX.

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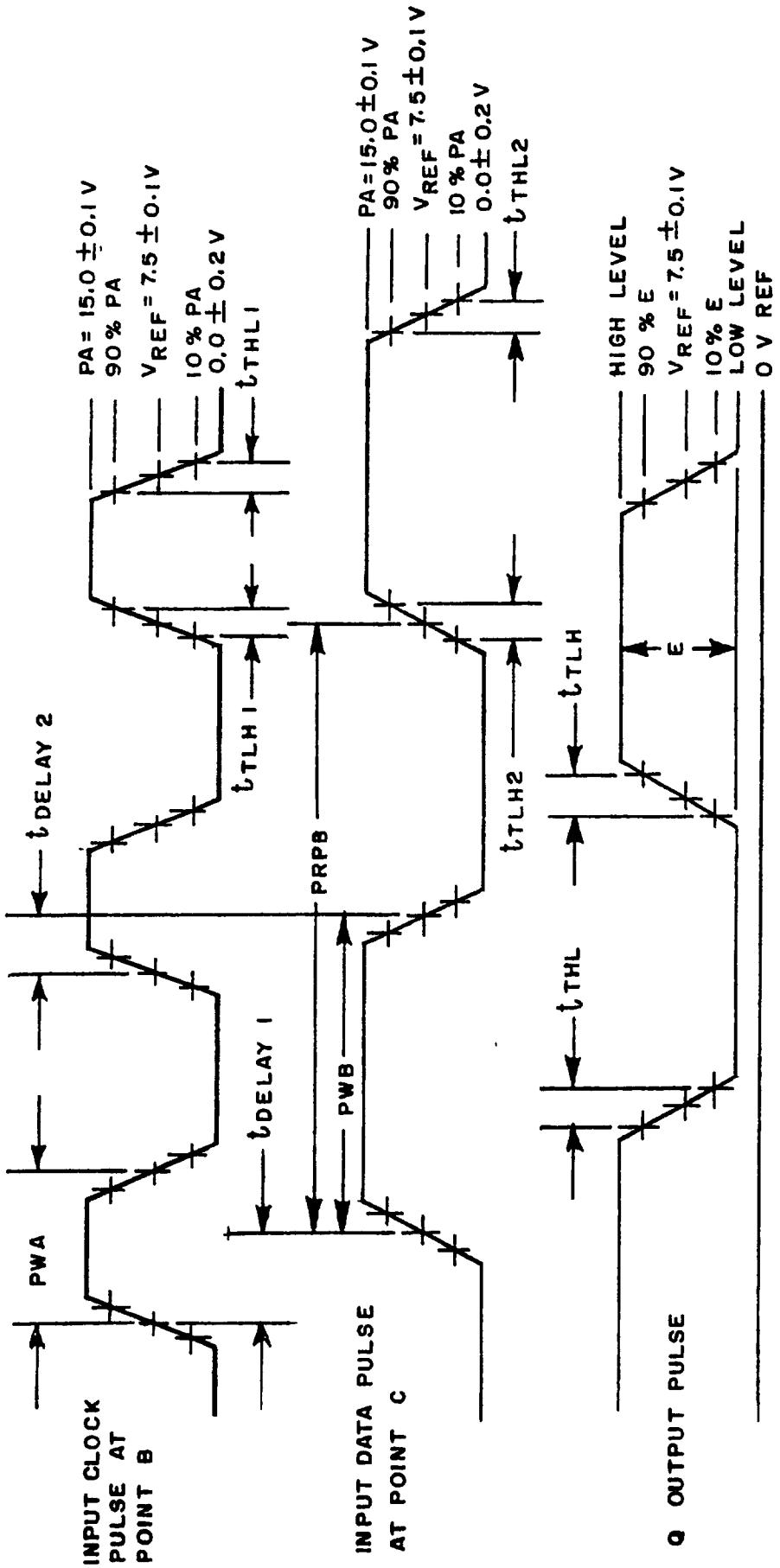


FIGURE 27. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXVII and XXVIII.

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NOTES:
1. Clock input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in external trigger mode.
2. Data input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 500 +50, -0 ns.
4. Before making measurements, verify that the output pulse high level is ≥ 13.5 V and that the low level is ≤ 1.5 V.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 27. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XXVII and XXVIII - Continued.

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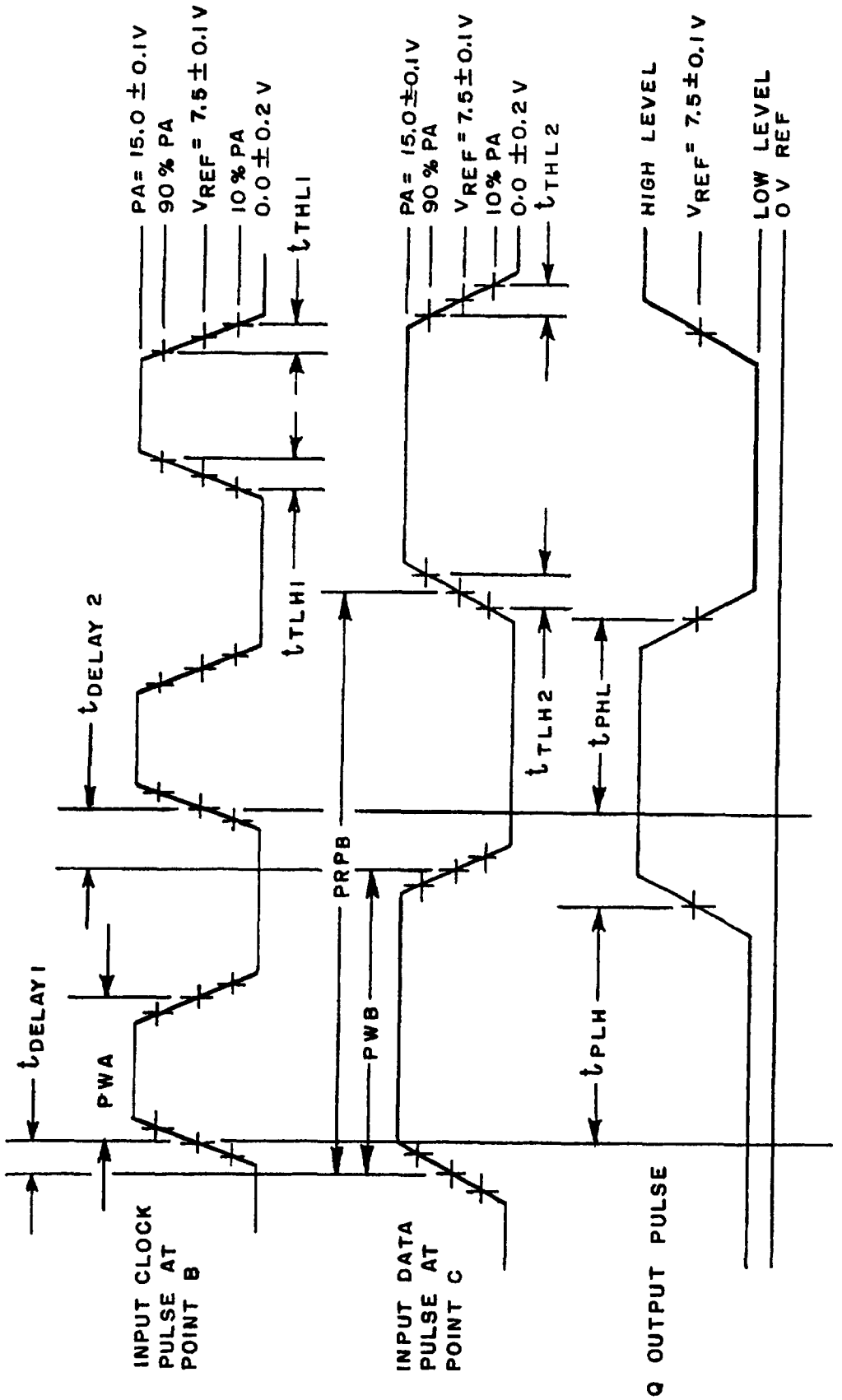


FIGURE 28. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIX and XXX.

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NOTES:
1. Clock input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   Put item J01-1 in external trigger mode.
2. Data input pulse parameters:
   PWB = 2.0 ±0.2 µs.
   PRPB = 4.0 ±0.4 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. $t_{DELAY1} = t_{DELAY2}$ = 500 +50, -0 ns.
4. Before making measurements, verify that the output pulse high level is ≥ 13.5 V and that the low level is ≤ 1.5 V.
5. The lead length of electrical conductors between resistors R1 and R2 and input pins, items Y01-1 and Y01-2 to input and output pins,and between load circuit and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

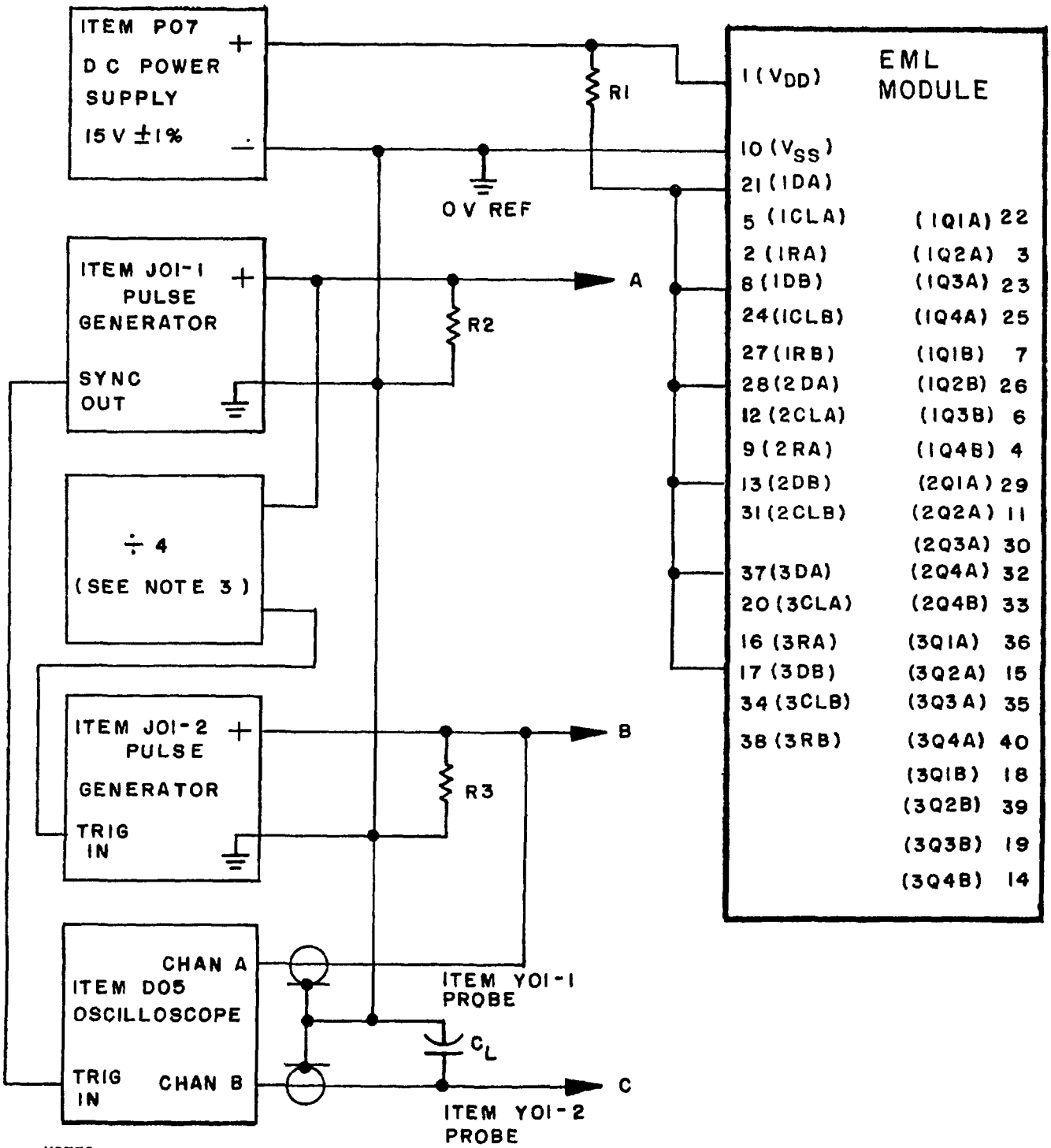
FIGURE 28. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIX and XXX - Continued.

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TABLE XXXI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
EE1	1RA to 1Q1A Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	5,24,27,12,9,31,20,16 34,38	2	22	17	170	ns	29,30
EE2	1RA to 1Q2A "	"	"	3	"	"	"	"
EE3	1RA to 1Q3A "	"	"	23	"	"	"	"
EE4	1RA to 1Q4A "	"	"	25	"	"	"	"
EE5	1RB to 1Q1B "	24,5,2,12,9,31,20,16, 34,38	27	7	"	"	"	"
EE6	1RB to 1Q2B "	"	"	26	"	"	"	"
EE7	1RB to 1Q3B "	"	"	6	"	"	"	"
EE8	1RB to 1Q4B "	"	"	4	"	"	"	"
EE9	2RA to 2Q1A "	12,5,2,24,27,31,20, 16,34,38	9	29	"	"	"	"
EE10	2RA to 2Q2A "	"	"	11	"	"	"	"
EE11	2RA to 2Q3A "	"	"	30	"	"	"	"
EE12	2RA to 2Q4A "	"	"	32	"	"	"	"
EE13	3RA to 3Q1A "	5,2,24,27,12,9,31, 34,38	16	36	"	"	"	"
EE14	3RA to 3Q2A "	"	"	15	"	"	"	"
EE15	3RA to 3Q3A "	"	"	35	"	"	"	"
EE16	3RA to 3Q4A "	"	"	40	"	"	"	"
EE17	3RB to 3Q1B "	5,2,24,27,12,9,31, 20,16	38	18	"	"	"	"
EE18	3RB to 3Q2B "	"	"	39	"	"	"	"
EE19	3RB to 3Q3B "	"	"	19	"	"	"	"
EE20	3RB to 3Q4B "	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).
3. The divide by 4 function may be either a commercially available unit (such as the Tektronix Model 00501) or any circuit that will provide one trigger input to item J01-2 for every fourth output pulse of item J01-1.

FIGURE 29. Propagation delay time (t_{PHL}) test circuit for table XXXI.

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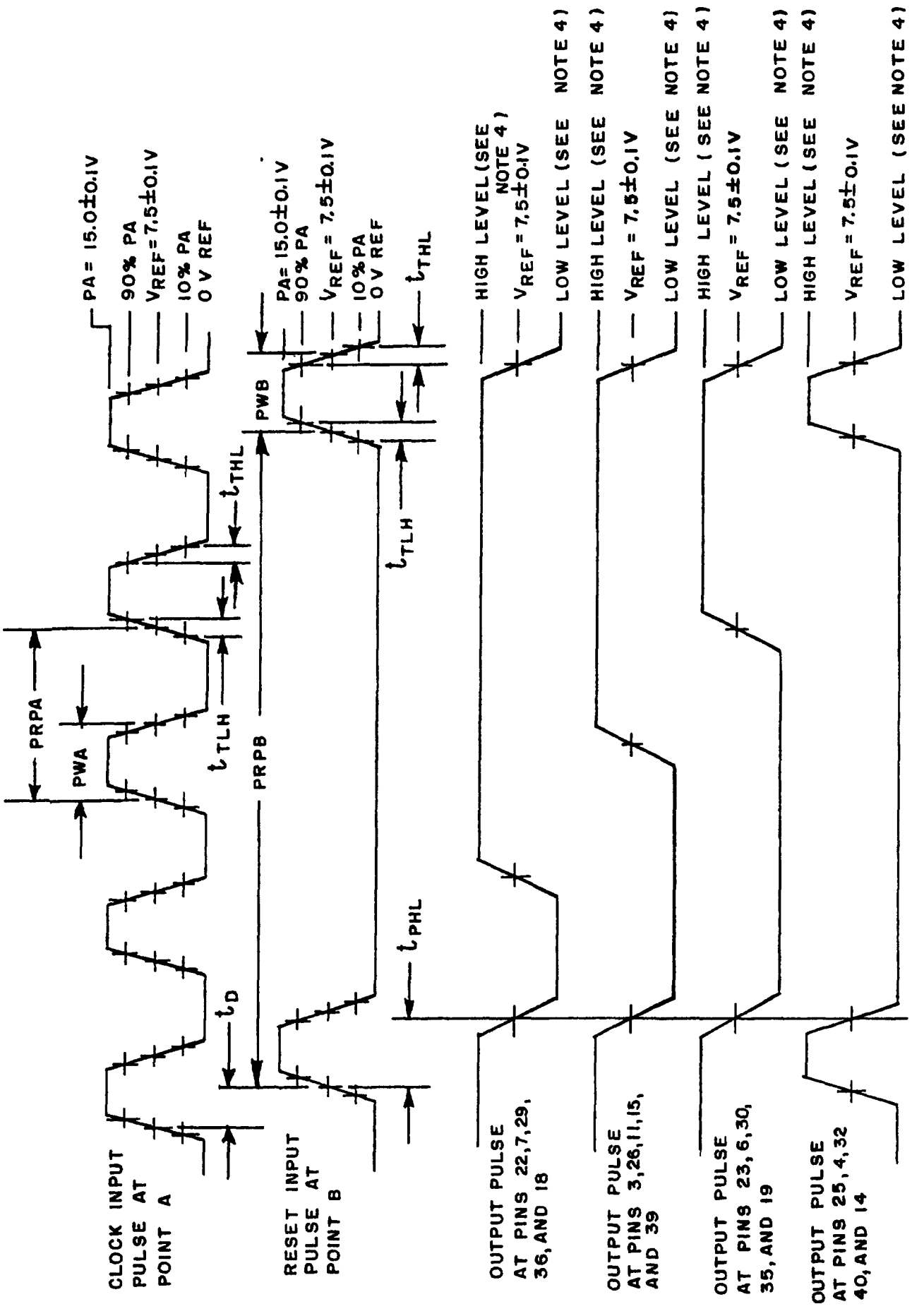


FIGURE 30. Propagation delay time (t_{PHL}) test circuit waveforms for table XXXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:20Z
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NOTES:
1. Item J01-1 input pulse parameters:
   PWA = 500 ±50 ns.
   PRPA = 2,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWB = 400 +8 ns, -0 ns.
   PRPB = 8,000 ±800 ns.
   t_D = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Before making measurements specified, verify that the output pulse high level is $\geq$ 13.8 V and the low level is $\leq$ 1.5 V.
4. Insure that the output pulse is $\geq$ 14 V for high level and $\leq$ 1.5 V for low level.

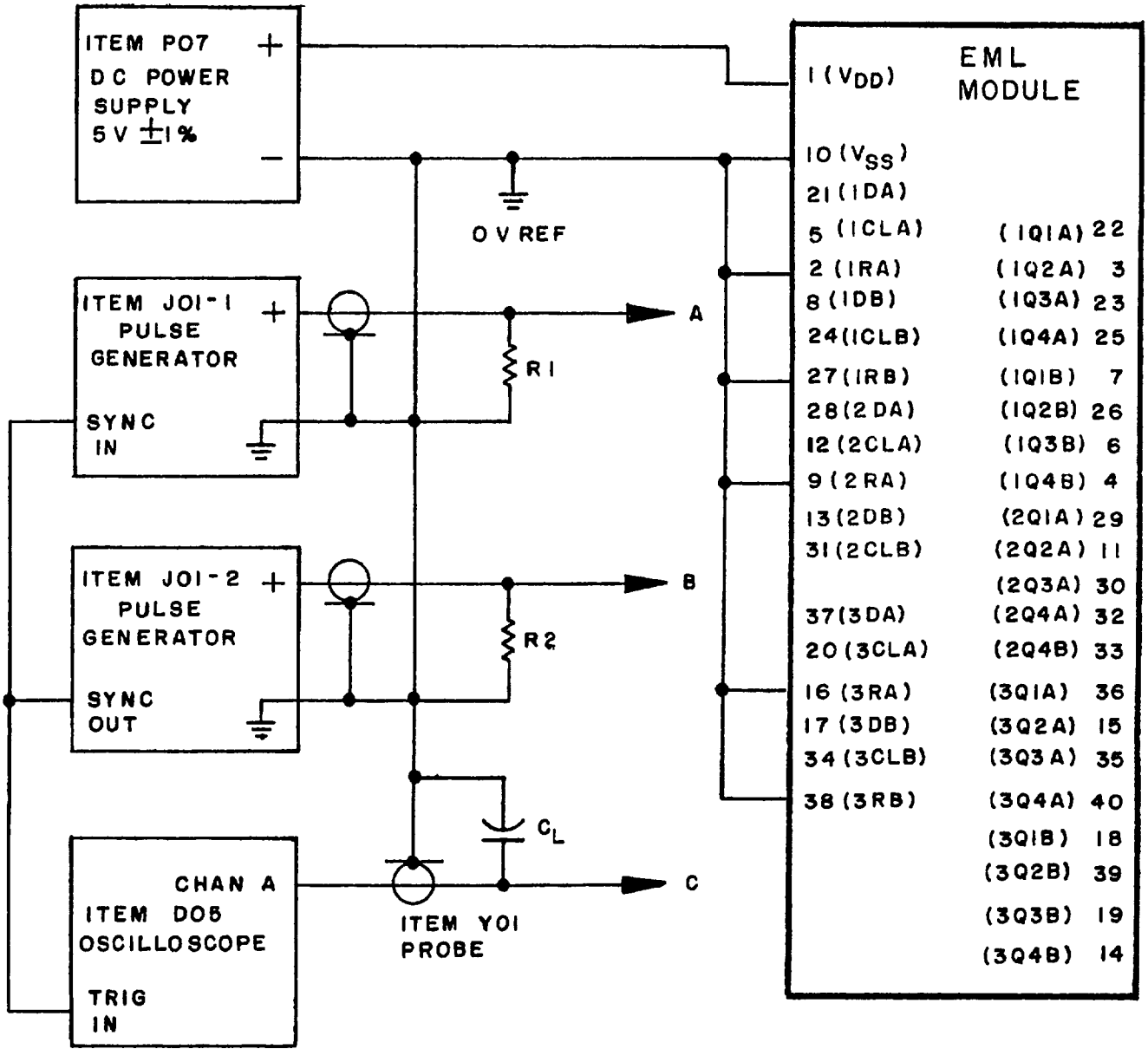
FIGURE 30. Propagation delay time (t_{PHL}) test circuit waveforms for table XXXI - Continued.

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TABLE XXXII. Setup time (t_{SL} and t_{SH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
FF1	1Q1A Setup time (t_{SL} and T_{SH})	5,24,12,31, 20,34	21,8,28,13, 37,17	22	2.3	3.7	µs	31,32
FF2	1Q2A "	"	"	3	"	"	"	"
FF3	1Q3A "	"	"	23	"	"	"	"
FF4	1Q4A "	"	"	25	"	"	"	"
FF5	1Q1B "	"	"	7	"	"	"	"
FF6	1Q2B "	"	"	26	"	"	"	"
FF7	1Q3B "	"	"	6	"	"	"	"
FF8	1Q4B "	"	"	4	"	"	"	"
FF9	2Q1A "	"	"	29	"	"	"	"
FF10	2Q2A "	"	"	11	"	"	"	"
FF11	2Q3A "	"	"	30	"	"	"	"
FF12	2Q4A "	"	"	32	"	"	"	"
FF13	2Q4B "	"	"	33	"	"	"	"
FF14	3Q1A "	"	"	36	"	"	"	"
FF15	3Q2A "	"	"	15	"	"	"	"
FF16	3Q3A "	"	"	35	"	"	"	"
FF17	3Q4A "	"	"	40	"	"	"	"
FF18	3Q1B "	"	"	18	"	"	"	"
FF19	3Q2B "	"	"	39	"	"	"	"
FF20	3Q3B "	"	"	19	"	"	"	"
FF21	3Q4B "	"	"	14	"	"	"	"

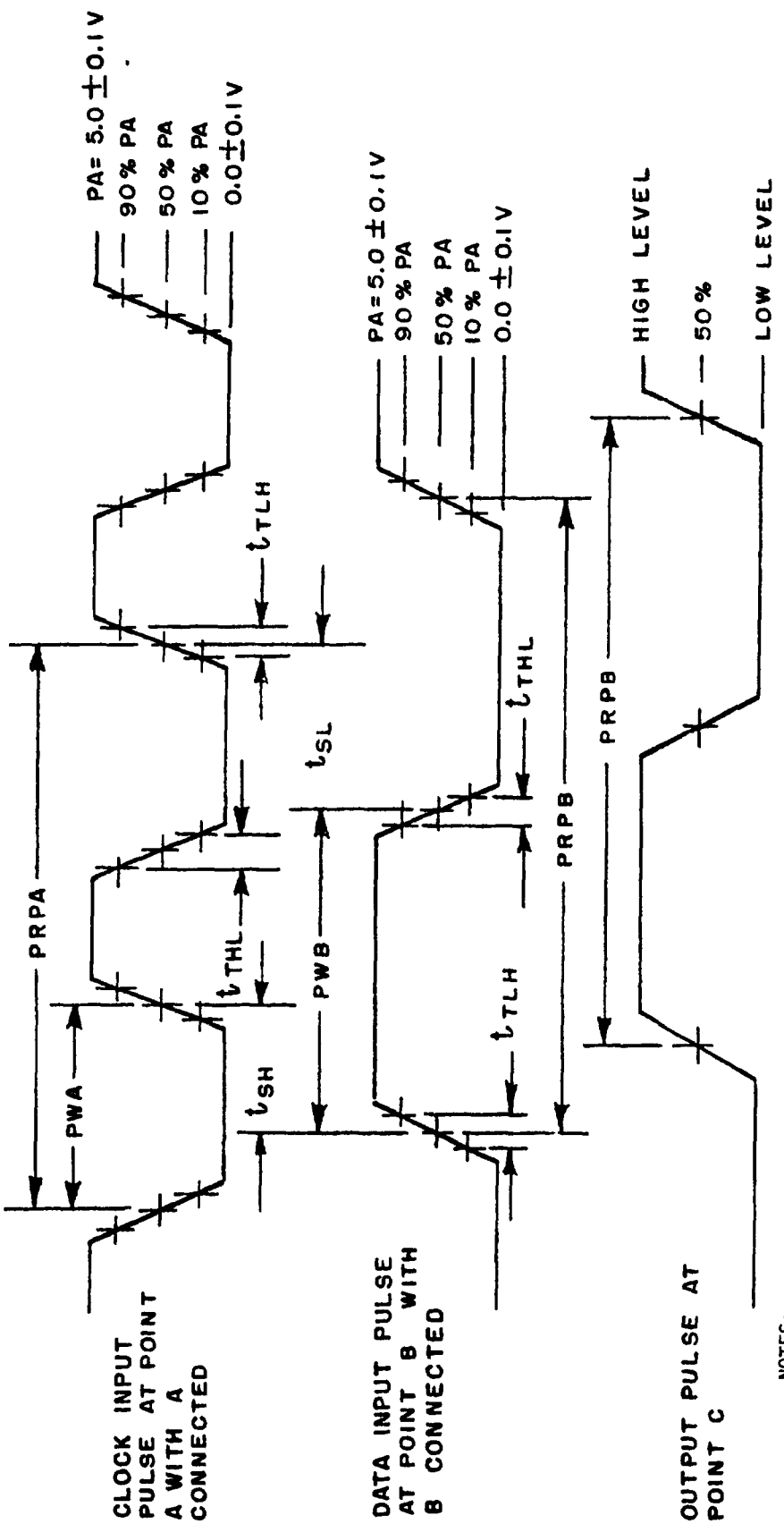
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:20Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 50Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 31. Setup time (t_{SL} and t_{SH}) test circuit for table XXXII.

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NOTES:
1. Item J01-1 input pulse parameters:
   PRPA = 1,500 ±150 ns.
   PWA = 750 ±75 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PRPB = 3,000 ±300 ns.
   PWB = 1,500 ±150 ns.
   $t_{SH} = t_{SL}$ = 350 +4, -0 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.

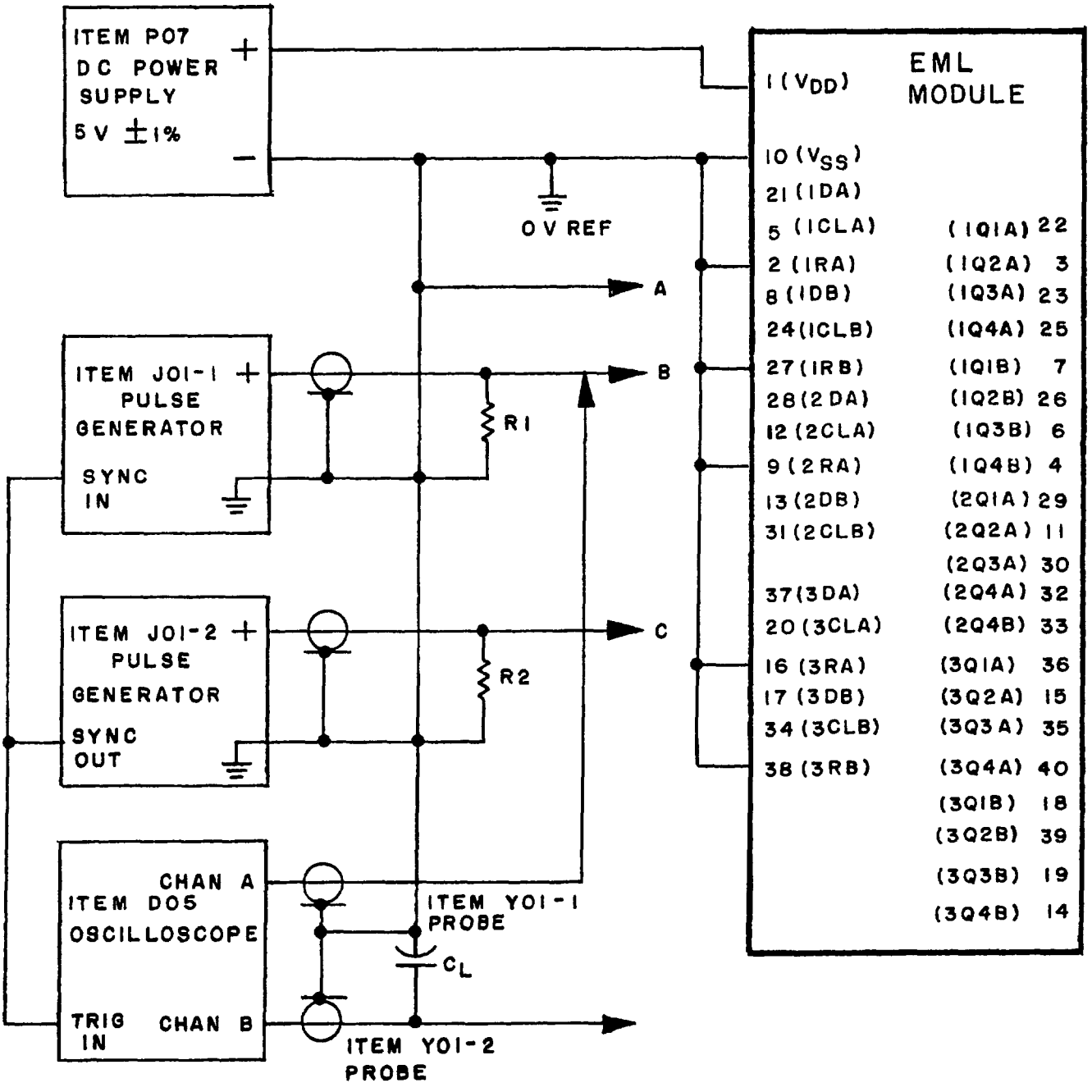
FIGURE 32. Setup time (t_{SL} and t_{SH}) test circuit waveforms for table XXXII.

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TABLE XXXIII. Maximum clock frequency check minimum pulse width test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
GG1	1Q1A	Maximum clock frequency check and minimum pulse width	8,24,28,12,13,31,37,20,17,34	5	21	22	1900	2300	ns	33,34
GG2	1Q2A	"	"	"	"	3	"	"	"	"
GG3	1Q3A	"	"	"	"	23	"	"	"	"
GG4	1Q4A	"	"	"	"	25	"	"	"	"
GG5	1Q1B	"	21,5,28,12,13,31,37,20,17,34	24	8	7	"	"	"	"
GG6	1Q2B	"	"	"	"	26	"	"	"	"
GG7	1Q3B	"	"	"	"	6	"	"	"	"
GG8	1Q4B	"	"	"	"	4	"	"	"	"
GG9	2Q1A	"	21,5,8,24,13,31,37,20,17,34	12	28	29	"	"	"	"
GG10	2Q2A	"	"	"	"	11	"	"	"	"
GG11	2Q3A	"	"	"	"	30	"	"	"	"
GG12	2Q4A	"	"	"	"	32	"	"	"	"
GG13	2Q4B	"	21,5,8,24,28,12,37,20,17,34	31	13	33	"	"	"	"
GG14	3Q1A	"	21,5,8,24,28,12,13,31,17,34	20	37	36	"	"	"	"
GG15	3Q2A	"	"	"	"	15	"	"	"	"
GG16	3Q3A	"	"	"	"	35	"	"	"	"
GG17	3Q4A	"	"	"	"	40	"	"	"	"
GG18	3Q1B	"	21,5,8,24,28,12,13,20,31	34	17	18	"	"	"	"
GG19	3Q2B	"	"	"	"	39	"	"	"	"
GG20	3Q3B	"	"	"	"	19	"	"	"	"
GG21	3Q4B	"	"	"	"	14	"	"	"	"

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 33. Maximum clock frequency check and minimum pulse width test circuit for table XXXIII.

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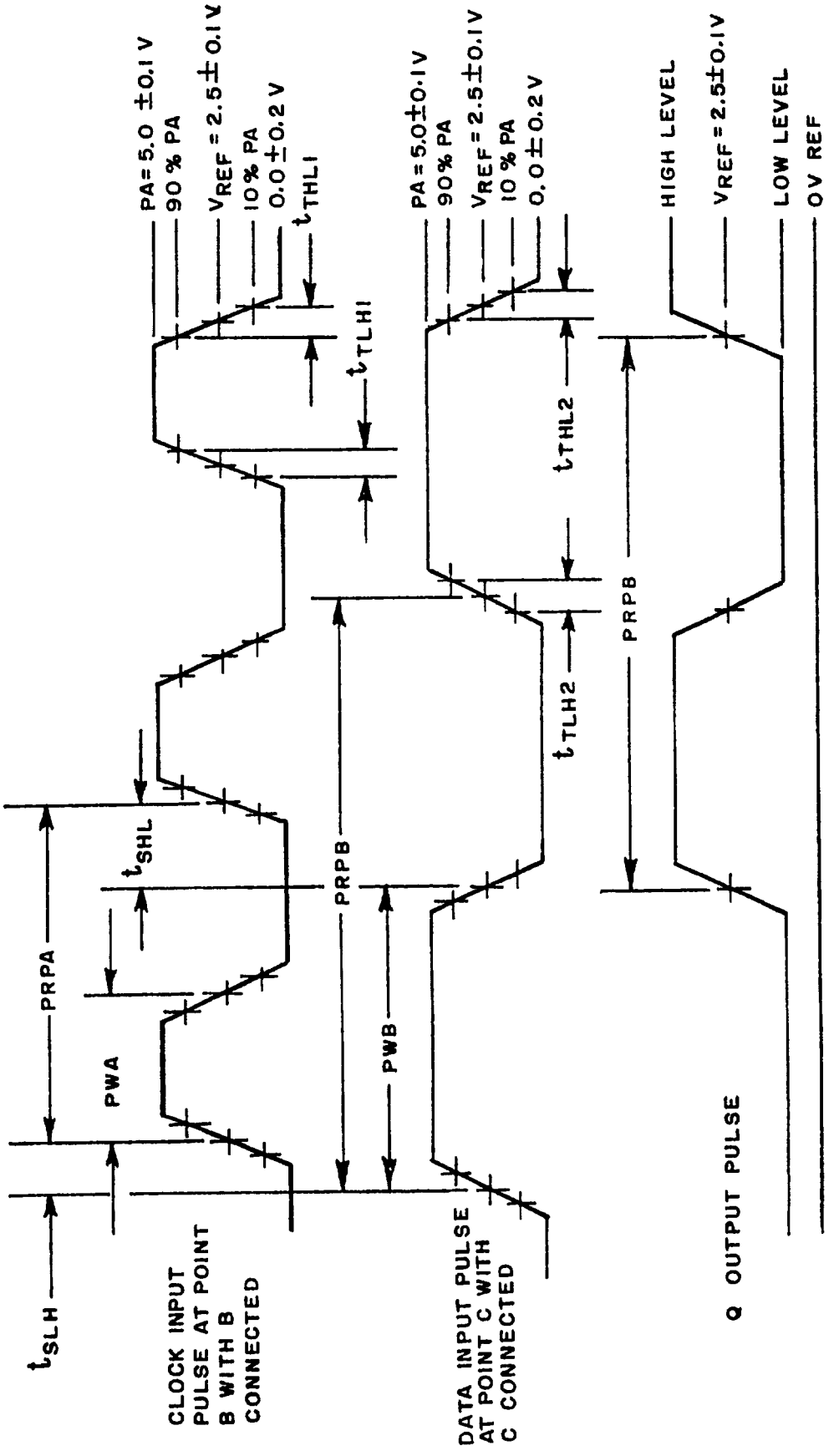


FIGURE 34. Maximum clock frequency check and minimum pulse width test circuit waveforms for table XXXIII.

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NOTES:
1. Item J01-1 input pulse parameters:
   PWA = 300 +30, -0 ns.
   PRPA = 1.05 ±0.05 µs.
   $t_{SLH} = t_{SHL}$ = 500 ±100 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWB = 1.05 ±0.05 µs.
   PRPB = 2.1 ±0.1 µs.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
3. Place item J01-1 in the double pulse mode and use the pulse width and delay adjustments of items J01-1 and J01-2 to obtain the parameters of notes 1 and 2.

FIGURE 34. Maximum clock frequency check and minimum pulse width test circuit waveforms for table XXXIII - Continued.

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TABLE XXXIV. Power supply filter capacitance.

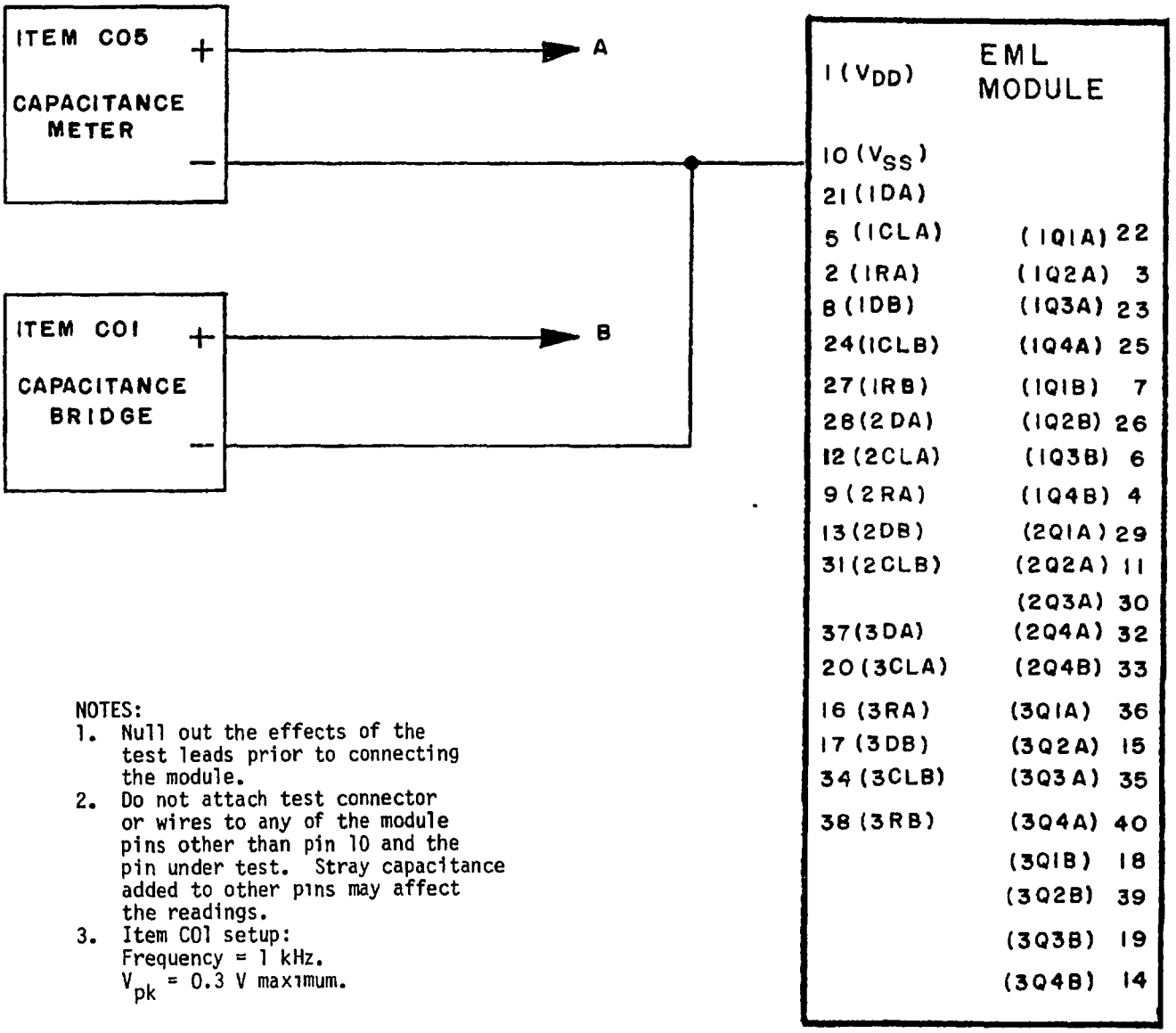
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
HH	Power supply filter capacitance	Open	1	0.007	0.013	µF	35

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TABLE XXXV. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
JJ1	1DA	Input capacitance	21	Open	2	15	pF	35
JJ2	1CLA	"	5	"	"	"	"	"
JJ3	1RA	"	2	"	"	"	"	"
JJ4	1DB	"	8	"	"	"	"	"
JJ5	1CLB	"	24	"	"	"	"	"
JJ6	1RB	"	27	"	"	"	"	"
JJ7	2DA	"	28	"	"	"	"	"
JJ8	2CLA	"	12	"	"	"	"	"
JJ9	2RA	"	9	"	"	"	"	"
JJ10	2DB	"	13	"	"	"	"	"
JJ11	2CLB	"	31	"	"	"	"	"
JJ12	3DA	"	37	"	"	"	"	"
JJ13	3CLA	"	20	"	"	"	"	"
JJ14	3RA	"	16	"	"	"	"	"
JJ15	3DB	"	17	"	"	"	"	"
JJ16	3CLB	"	34	"	"	"	"	"
JJ17	3RB	"	38	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 35. Power supply filter and input capacitance test circuit for tables XXXIV and XXXV.

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